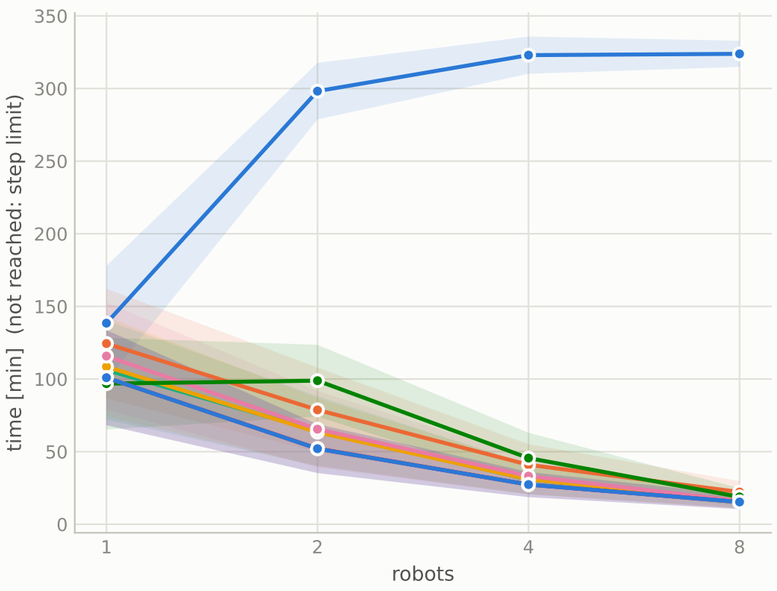
Multi-agent coverage, paper: method comparison

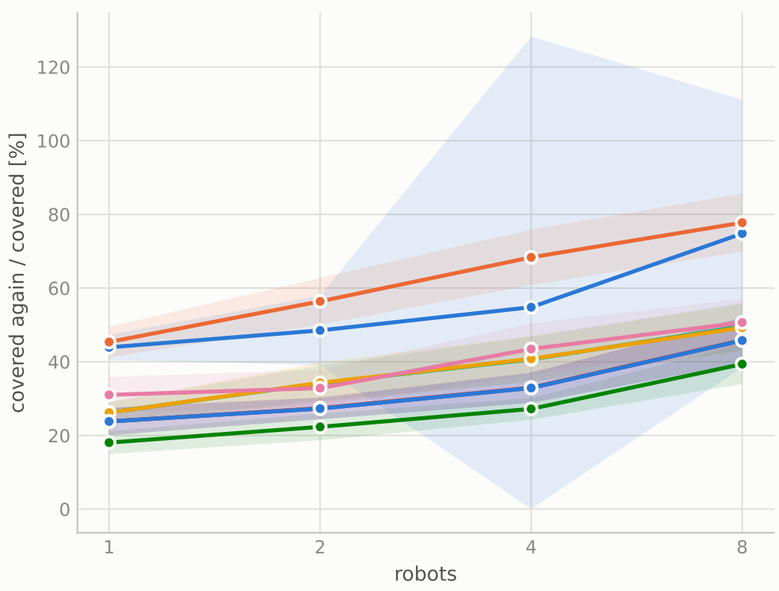
38 maps x 3 episode(s) x 1 training seed(s) per method and team size, deterministic actions, same start poses for every policy. Mean over maps of the mean over seeds, 95 % CI over seeds and maps.

Single-agent xN · Single-agent xN + flip · MAPPO · MAPPO + flip · MAPPO (last) · Single-agent fine-tuned xN + flip · IPPO · IPPO + flip · IPPO (last)

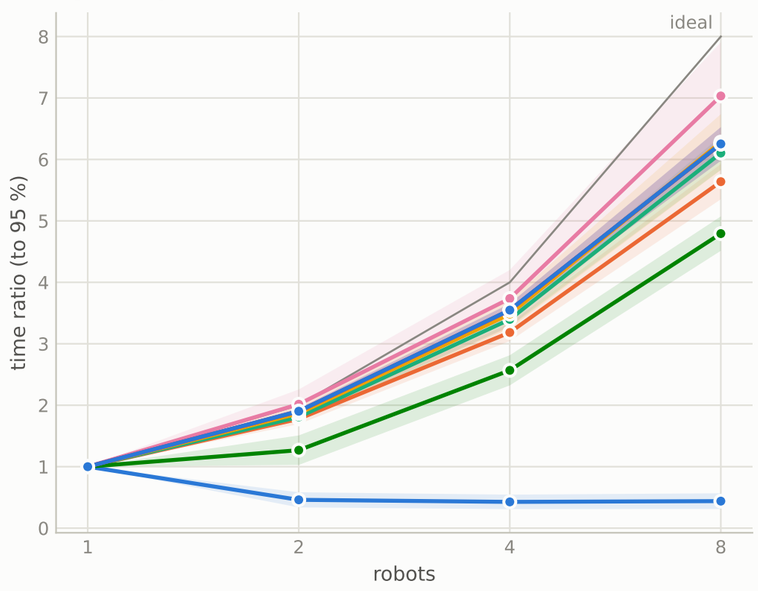
Time to 95 % coverage



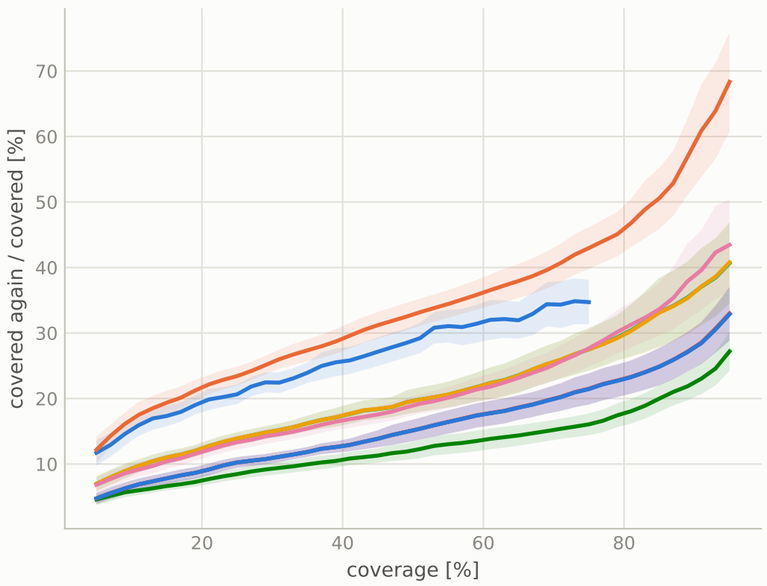
Overlap when 95 % coverage is reached



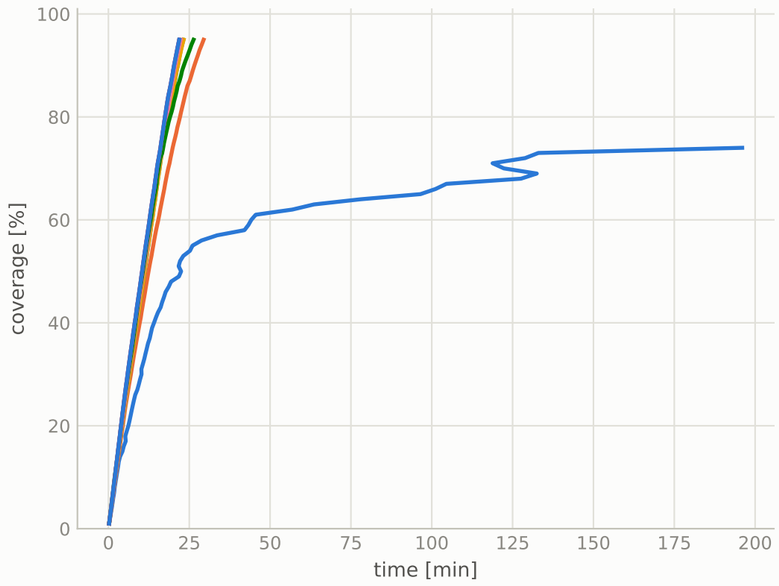
Speedup over 1 robot



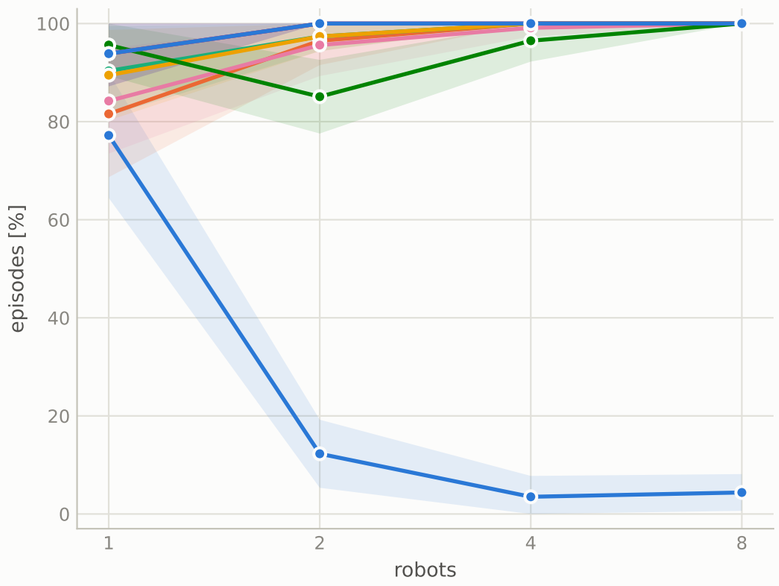
Overlap while covering, 4 robots



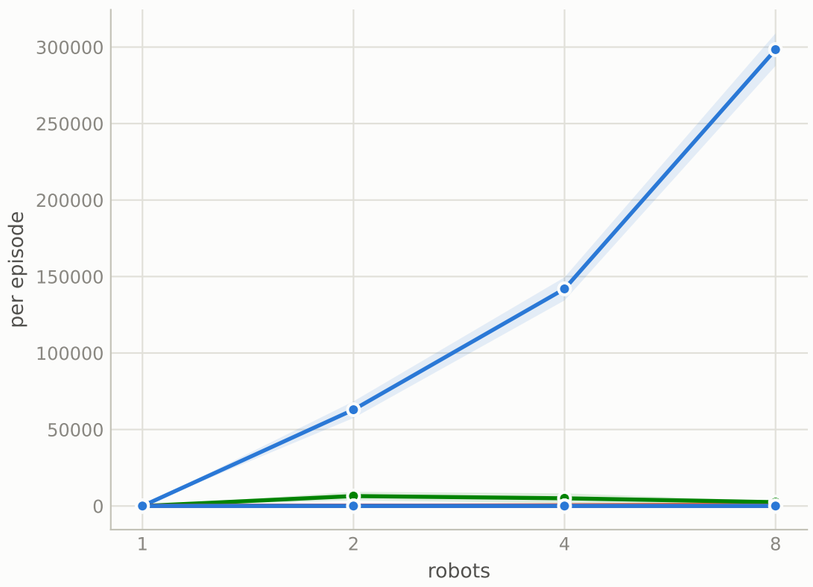
Coverage over time, 4 robots (median map)



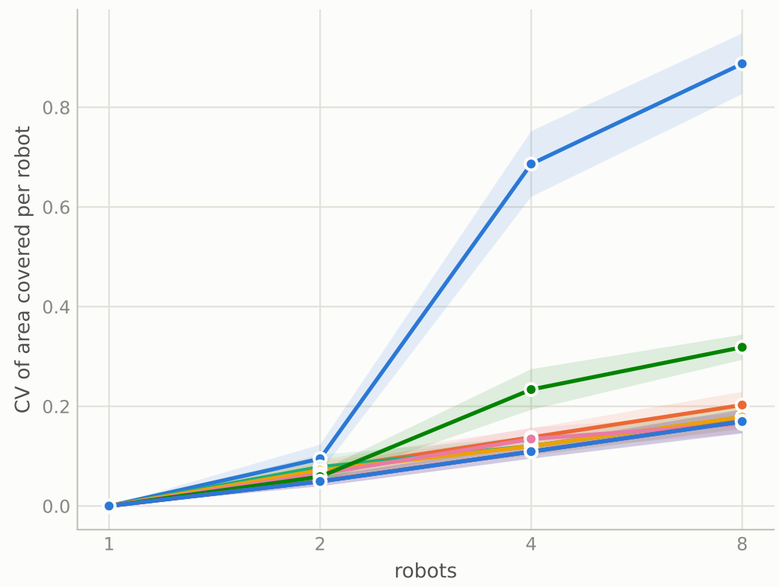
Success rate (95 % coverage)



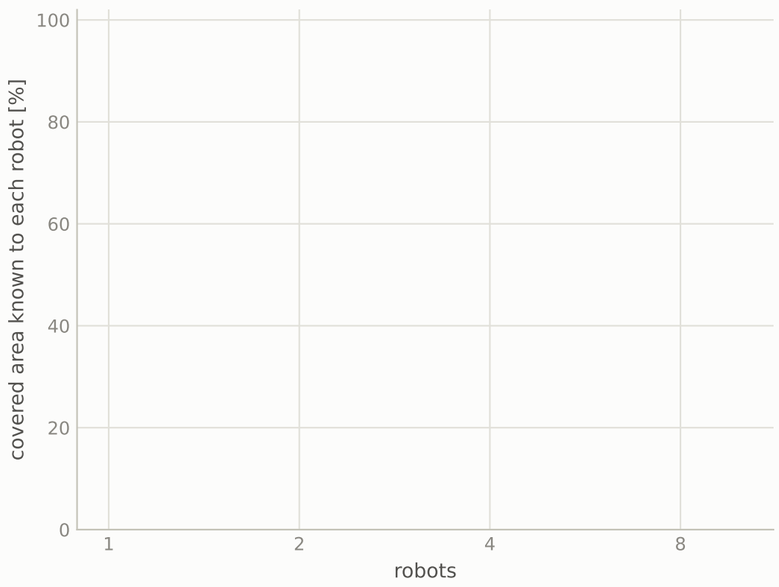
Collisions between robots



Workload imbalance



Belief recall (limited communication)



Summary, 4 robots: mean [95 % CI]

Method	Comm.	Seeds	Success (95 %)	Time to 95 %	Overlap at 95 %	Robot coll.	Imbalance	Belief recall	Connectivity
Single-agent xN	unlimited	1	4 [0, 8] %	323.0 [310.2, 335.8] mir	54.8 [0.0, 128.3] %	141926.7 [134379.8, 14	0.69 [0.62, 0.75]	100 [100, 100] %	100 [100, 100] %
Single-agent xN + flip	unlimited	1	100 [100, 100] %	41.0 [26.7, 55.2] min	68.3 [60.8, 75.9] %	504.2 [409.4, 599.1]	0.14 [0.12, 0.16]	100 [100, 100] %	100 [100, 100] %
MAPPO	unlimited	1	100 [100, 100] %	31.0 [20.7, 41.4] min	40.7 [34.5, 46.9] %	0.5 [0.1, 1.0]	0.12 [0.10, 0.14]	100 [100, 100] %	100 [100, 100] %
MAPPO + flip	unlimited	1	100 [100, 100] %	31.1 [20.7, 41.4] min	40.8 [34.6, 47.0] %	0.5 [0.1, 1.0]	0.12 [0.10, 0.14]	100 [100, 100] %	100 [100, 100] %
MAPPO (last)	unlimited	1	99 [97, 100] %	33.2 [20.6, 45.8] min	43.5 [36.5, 50.4] %	2.4 [1.1, 3.8]	0.13 [0.11, 0.16]	100 [100, 100] %	100 [100, 100] %
Single-agent fine-tuned xN + flip	unlimited	1	96 [92, 100] %	45.6 [28.1, 63.0] min	27.2 [24.2, 30.2] %	5034.2 [1776.2, 8292.1]	0.23 [0.19, 0.27]	100 [100, 100] %	100 [100, 100] %
IPPO	unlimited	1	100 [100, 100] %	27.4 [18.7, 36.0] min	32.9 [28.8, 37.0] %	0.2 [0.1, 0.3]	0.11 [0.09, 0.12]	100 [100, 100] %	100 [100, 100] %

Multi-agent coverage, paper: statistical tests

38 maps x 3 episode(s) x 1 training seed(s) per method and team size, deterministic actions, same start poses for every policy. Mean over maps of the mean over seeds, 95 % CI over seeds and maps.

Single-agent xN
Single-agent xN + flip
MAPPO
MAPPO + flip
MAPPO (last)
Single-agent fine-tuned xN + flip
IPPO
IPPO + flip
IPPO (last)

Each cell: difference B - A [95 % CI], Holm-corrected p-value (within each metric), and P(B better): probability that B beats A on a random map, for a random seed of each (0.5 = no difference).
Bold: significant at the 5 % level after the correction. A single seed gives no p-value. Policies evaluated twice (same checkpoints) are paired by seed.

Time to the target coverage (95 %): relative difference, lower is better

B vs A	1 robots	2 robots	4 robots	8 robots
Single-agent xN + flip vs Single-agent x	-10.1 % [-17.6, -2.7] p n/a P(B better) 0.61	-73.6 % [-83.6, -63.6] p n/a P(B better) 1.00	-87.3 % [-92.8, -81.8] p n/a P(B better) 1.00	-93.1 % [-97.1, -89.2] p n/a P(B better) 1.00
MAPPO vs Single-agent xN	-23.3 % [-33.3, -13.3] p n/a P(B better) 0.93	-78.5 % [-87.3, -69.7] p n/a P(B better) 1.00	-90.4 % [-95.1, -85.7] p n/a P(B better) 1.00	-95.0 % [-98.4, -91.6] p n/a P(B better) 1.00
MAPPO + flip vs Single-agent xN	-21.6 % [-32.2, -11.0] p n/a P(B better) 0.91	-78.7 % [-87.4, -70.1] p n/a P(B better) 1.00	-90.4 % [-95.1, -85.7] p n/a P(B better) 1.00	-95.0 % [-98.4, -91.6] p n/a P(B better) 1.00
MAPPO (last) vs Single-agent xN	-16.4 % [-26.3, -6.4] p n/a P(B better) 0.92	-78.0 % [-87.5, -68.6] p n/a P(B better) 0.99	-89.7 % [-94.9, -84.6] p n/a P(B better) 1.00	-95.1 % [-98.4, -91.7] p n/a P(B better) 1.00
Single-agent fine-tuned xN + flip vs Sin	-30.1 % [-40.6, -19.5] p n/a P(B better) 0.99	-66.8 % [-78.3, -55.4] p n/a P(B better) 0.95	-85.9 % [-92.1, -79.7] p n/a P(B better) 1.00	-94.2 % [-97.7, -90.7] p n/a P(B better) 1.00
IPPO vs Single-agent xN	-27.1 % [-37.8, -16.4] p n/a P(B better) 0.99	-82.5 % [-89.8, -75.3] p n/a P(B better) 1.00	-91.5 % [-96.0, -87.1] p n/a P(B better) 1.00	-95.3 % [-98.6, -91.9] p n/a P(B better) 1.00
IPPO + flip vs Single-agent xN	-27.1 % [-37.8, -16.4] p n/a P(B better) 0.99	-82.5 % [-89.8, -75.2] p n/a P(B better) 1.00	-91.5 % [-96.0, -87.1] p n/a P(B better) 1.00	-95.3 % [-98.6, -91.9] p n/a P(B better) 1.00
IPPO (last) vs Single-agent xN	-27.1 % [-37.8, -16.4] p n/a P(B better) 0.99	-82.5 % [-89.8, -75.3] p n/a P(B better) 1.00	-91.5 % [-96.0, -87.1] p n/a P(B better) 1.00	-95.3 % [-98.6, -91.9] p n/a P(B better) 1.00
MAPPO vs Single-agent xN + flip	-14.7 % [-20.1, -9.2] p n/a P(B better) 0.96	-18.5 % [-27.2, -9.8] p n/a P(B better) 0.92	-24.2 % [-35.7, -12.8] p n/a P(B better) 0.97	-27.0 % [-38.5, -15.5] p n/a P(B better) 0.97
MAPPO + flip vs Single-agent xN + flip	-12.8 % [-19.2, -6.4] p n/a P(B better) 0.93	-19.5 % [-28.6, -10.4] p n/a P(B better) 0.92	-24.2 % [-35.6, -12.8] p n/a P(B better) 0.97	-27.3 % [-39.0, -15.6] p n/a P(B better) 0.97
MAPPO (last) vs Single-agent xN + flip	-6.9 % [-12.8, -1.1] p n/a P(B better) 0.87	-16.9 % [-26.4, -7.3] p n/a P(B better) 0.92	-19.0 % [-31.1, -6.9] p n/a P(B better) 0.95	-28.0 % [-39.4, -16.6] p n/a P(B better) 1.00
Single-agent fine-tuned xN + flip vs Sin	-22.2 % [-29.7, -14.7] p n/a P(B better) 0.99	+25.5 % [-9.5, +60.6] p n/a P(B better) 0.61	+11.3 % [-14.3, +36.9] p n/a P(B better) 0.71	-15.2 % [-24.4, -6.0] p n/a P(B better) 0.87
IPPO vs Single-agent xN + flip	-18.9 % [-25.6, -12.2] p n/a P(B better) 0.99	-33.9 % [-51.7, -16.1] p n/a P(B better) 1.00	-33.2 % [-47.6, -18.8] p n/a P(B better) 1.00	-30.9 % [-43.2, -18.6] p n/a P(B better) 1.00
IPPO + flip vs Single-agent xN + flip	-18.9 % [-25.6, -12.2] p n/a P(B better) 0.99	-33.7 % [-51.4, -16.1] p n/a P(B better) 1.00	-33.1 % [-47.5, -18.7] p n/a P(B better) 1.00	-30.9 % [-43.2, -18.6] p n/a P(B better) 1.00
IPPO (last) vs Single-agent xN + flip	-19.0 % [-25.6, -12.3] p n/a P(B better) 0.99	-33.9 % [-51.7, -16.1] p n/a P(B better) 1.00	-33.2 % [-47.6, -18.8] p n/a P(B better) 1.00	-30.9 % [-43.2, -18.6] p n/a P(B better) 1.00

Overlap when the target coverage is reached (95 %): percentage points, lower is better

B vs A	1 robots	2 robots	4 robots	8 robots
Single-agent xN + flip vs Single-agent x	+1.5 pp [-0.0, +2.9] p n/a P(B better) 0.32	+7.9 pp [+3.4, +12.3] p n/a P(B better) 0.36	+13.5 pp [-37.5, +64.5] p n/a P(B better) 0.33	+2.9 pp [-46.5, +52.4] p n/a P(B better) 0.00
MAPPO vs Single-agent xN	-17.7 pp [-20.5, -14.8] p n/a P(B better) 0.97	-14.3 pp [-22.3, -6.4] p n/a P(B better) 0.91	-14.2 pp [-41.7, +13.4] p n/a P(B better) 1.00	-25.2 pp [-59.5, +9.1] p n/a P(B better) 0.80
MAPPO + flip vs Single-agent xN	-17.7 pp [-20.4, -15.0] p n/a P(B better) 0.97	-14.2 pp [-21.6, -6.9] p n/a P(B better) 1.00	-14.0 pp [-42.2, +14.1] p n/a P(B better) 1.00	-25.5 pp [-59.8, +8.8] p n/a P(B better) 0.80
MAPPO (last) vs Single-agent xN	-12.9 pp [-16.7, -9.1] p n/a P(B better) 0.88	-15.7 pp [-28.2, -3.1] p n/a P(B better) 0.82	-11.3 pp [-53.9, +31.2] p n/a P(B better) 1.00	-24.1 pp [-55.3, +7.0] p n/a P(B better) 0.80
Single-agent fine-tuned xN + flip vs Sin	-25.9 pp [-28.6, -23.1] p n/a P(B better) 1.00	-26.2 pp [-33.2, -19.1] p n/a P(B better) 1.00	-27.6 pp [-66.8, +11.6] p n/a P(B better) 1.00	-35.4 pp [-60.6, -10.2] p n/a P(B better) 1.00
IPPO vs Single-agent xN	-20.1 pp [-22.9, -17.4] p n/a P(B better) 0.97	-21.2 pp [-25.9, -16.6] p n/a P(B better) 1.00	-21.9 pp [-66.5, +22.7] p n/a P(B better) 1.00	-29.0 pp [-48.7, -9.3] p n/a P(B better) 1.00
IPPO + flip vs Single-agent xN	-20.1 pp [-22.9, -17.4] p n/a P(B better) 0.97	-21.1 pp [-25.8, -16.4] p n/a P(B better) 1.00	-21.8 pp [-66.4, +22.8] p n/a P(B better) 1.00	-28.9 pp [-48.7, -9.2] p n/a P(B better) 1.00
IPPO (last) vs Single-agent xN	-20.1 pp [-22.9, -17.4] p n/a P(B better) 0.97	-21.2 pp [-25.9, -16.6] p n/a P(B better) 1.00	-21.9 pp [-66.5, +22.7] p n/a P(B better) 1.00	-29.0 pp [-48.7, -9.3] p n/a P(B better) 1.00
MAPPO vs Single-agent xN + flip	-19.1 pp [-22.6, -15.7] p n/a P(B better) 1.00	-22.2 pp [-26.3, -18.2] p n/a P(B better) 0.97	-27.7 pp [-33.9, -21.5] p n/a P(B better) 0.97	-28.2 pp [-35.9, -20.4] p n/a P(B better) 0.92
MAPPO + flip vs Single-agent xN + flip	-19.2 pp [-22.5, -15.8] p n/a P(B better) 1.00	-22.1 pp [-26.2, -18.0] p n/a P(B better) 1.00	-27.5 pp [-33.7, -21.4] p n/a P(B better) 0.97	-28.5 pp [-36.4, -20.6] p n/a P(B better) 0.92
MAPPO (last) vs Single-agent xN + flip	-14.4 pp [-18.5, -10.2] p n/a P(B better) 0.87	-23.5 pp [-30.0, -17.1] p n/a P(B better) 0.89	-24.9 pp [-29.6, -20.1] p n/a P(B better) 0.97	-27.1 pp [-34.7, -19.5] p n/a P(B better) 0.95
Single-agent fine-tuned xN + flip vs Sin	-27.3 pp [-30.2, -24.4] p n/a P(B better) 1.00	-34.1 pp [-39.9, -28.2] p n/a P(B better) 1.00	-41.1 pp [-47.9, -34.4] p n/a P(B better) 1.00	-38.4 pp [-45.9, -30.9] p n/a P(B better) 1.00
IPPO vs Single-agent xN + flip	-21.6 pp [-25.1, -18.0] p n/a P(B better) 0.97	-29.1 pp [-34.7, -23.5] p n/a P(B better) 1.00	-35.5 pp [-42.1, -28.8] p n/a P(B better) 1.00	-32.0 pp [-39.5, -24.5] p n/a P(B better) 0.97
IPPO + flip vs Single-agent xN + flip	-21.6 pp [-25.1, -18.0] p n/a P(B better) 0.97	-29.0 pp [-34.5, -23.5] p n/a P(B better) 1.00	-35.3 pp [-42.0, -28.7] p n/a P(B better) 1.00	-31.9 pp [-39.4, -24.4] p n/a P(B better) 0.97
IPPO (last) vs Single-agent xN + flip	-21.6 pp [-25.1, -18.0] p n/a P(B better) 0.97	-30.1 pp [-34.7, -23.5] p n/a P(B better) 1.00	-35.5 pp [-42.1, -28.8] p n/a P(B better) 1.00	-32.0 pp [-39.5, -24.5] p n/a P(B better) 0.97

Success rate (95 %): percentage points, higher is better

B vs A	1 robots	2 robots	4 robots	8 robots
Single-agent xN + flip vs Single-agent x	+4.4 pp [-1.4, +10.2] p n/a P(B better) 0.54	+84.2 pp [+76.3, +92.2] p n/a P(B better) 1.00	+96.5 pp [+92.2, +100.7] p n/a P(B better) 1.00	+95.6 pp [+91.9, +99.4] p n/a P(B better) 1.00
MAPPO vs Single-agent xN	+13.2 pp [+3.8, +22.5] p n/a P(B better) 0.61	+85.1 pp [+78.0, +92.2] p n/a P(B better) 1.00	+96.5 pp [+92.2, +100.7] p n/a P(B better) 1.00	+95.6 pp [+91.9, +99.4] p n/a P(B better) 1.00
MAPPO + flip vs Single-agent xN	+12.3 pp [+2.6, +22.0] p n/a P(B better) 0.59	+85.1 pp [+78.0, +92.2] p n/a P(B better) 1.00	+96.5 pp [+92.2, +100.7] p n/a P(B better) 1.00	+95.6 pp [+91.9, +99.4] p n/a P(B better) 1.00
MAPPO (last) vs Single-agent xN	+7.0 pp [+0.2, +13.8] p n/a P(B better) 0.57	+83.3 pp [+74.6, +92.1] p n/a P(B better) 0.99	+95.6 pp [+91.1, +100.2] p n/a P(B better) 1.00	+95.6 pp [+91.9, +99.4] p n/a P(B better) 1.00
Single-agent fine-tuned xN + flip vs Sin	+18.4 pp [+7.1, +29.7] p n/a P(B better) 0.63	+72.8 pp [+60.9, +84.7] p n/a P(B better) 0.95	+93.0 pp [+87.2, +98.8] p n/a P(B better) 1.00	+95.6 pp [+91.9, +99.4] p n/a P(B better) 1.00
IPPO vs Single-agent xN	+16.7 pp [+6.6, +26.8] p n/a P(B better) 0.63	+87.7 pp [+80.8, +94.7] p n/a P(B better) 1.00	+96.5 pp [+92.2, +100.7] p n/a P(B better) 1.00	+95.6 pp [+91.9, +99.4] p n/a P(B better) 1.00
IPPO + flip vs Single-agent xN	+16.7 pp [+6.6, +26.8] p n/a P(B better) 0.63	+87.7 pp [+80.8, +94.7] p n/a P(B better) 1.00	+96.5 pp [+92.2, +100.7] p n/a P(B better) 1.00	+95.6 pp [+91.9, +99.4] p n/a P(B better) 1.00
IPPO (last) vs Single-agent xN	+16.7 pp [+6.6, +26.8] p n/a P(B better) 0.63	+87.7 pp [+80.8, +94.7] p n/a P(B better) 1.00	+96.5 pp [+92.2, +100.7] p n/a P(B better) 1.00	+95.6 pp [+91.9, +99.4] p n/a P(B better) 1.00
MAPPO vs Single-agent xN + flip	+8.8 pp [+0.1, +17.5] p n/a P(B better) 0.55	+0.9 pp [-2.2, +4.0] p n/a P(B better) 0.51	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50
MAPPO + flip vs Single-agent xN + flip	+7.9 pp [-1.1, +16.9] p n/a P(B better) 0.54	+0.9 pp [-2.2, +4.0] p n/a P(B better) 0.51	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50
MAPPO (last) vs Single-agent xN + flip	+2.6 pp [-4.7, +10.0] p n/a P(B better) 0.51	-0.9 pp [-2.7, +0.9] p n/a P(B better) 0.49	-0.9 pp [-2.7, +0.9] p n/a P(B better) 0.49	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50
Single-agent fine-tuned xN + flip vs Sin	+14.0 pp [+2.7, +25.3] p n/a P(B better) 0.58	-11.4 pp [-21.0, -1.8] p n/a P(B better) 0.36	-3.5 pp [-7.8, +0.7] p n/a P(B better) 0.46	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50
IPPO vs Single-agent xN + flip	+12.3 pp [+2.3, +22.3] p n/a P(B better) 0.58	+3.5 pp [-1.4, +8.5] p n/a P(B better) 0.53	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50
IPPO + flip vs Single-agent xN + flip	+12.3 pp [+2.3, +22.3] p n/a P(B better) 0.58	+3.5 pp [-1.4, +8.5] p n/a P(B better) 0.53	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50
IPPO (last) vs Single-agent xN + flip	+12.3 pp [+2.3, +22.3] p n/a P(B better) 0.58	+3.5 pp [-1.4, +8.5] p n/a P(B better) 0.53	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50

Collisions between robots per episode: difference, lower is better

B vs A	1 robots	2 robots	4 robots	8 robots
Single-agent xN + flip vs Single-agent x	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-62753.88 [-68200.86, -57306.90] p n/a P(B better	-141422.47 [-148996.63, -133848.31] p n/a P(B be	-297329.46 [-308022.81, -286636.12] p n/a P(B better) 1
MAPPO vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-62942.37 [-68388.43, -57496.30] p n/a P(B better	-141926.19 [-149473.10, -134379.29] p n/a P(B be	-298332.05 [-308946.09, -287718.02] p n/a P(B better) 1
MAPPO + flip vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-62942.37 [-68388.43, -57496.30] p n/a P(B better	-141926.18 [-149473.09, -134379.28] p n/a P(B be	-298333.04 [-308947.32, -287718.77] p n/a P(B better) 1
MAPPO (last) vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-62927.67 [-68375.55, -57479.78] p n/a P(B better	-141924.28 [-149471.25, -134377.31] p n/a P(B be	-298325.65 [-308941.36, -287709.94] p n/a P(B better) 1
Single-agent fine-tuned xN + flip vs Sin	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-56442.46 [-63090.56, -49794.37] p n/a P(B better	-136892.56 [-145196.81, -128588.32] p n/a P(B be	-295832.46 [-306778.54, -284886.37] p n/a P(B better) 1
IPPO vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-62942.37 [-68388.43, -57496.30] p n/a P(B better	-141926.54 [-149473.48, -134379.60] p n/a P(B be	-298334.58 [-308949.01, -287720.14] p n/a P(B better) 1
IPPO + flip vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-62942.37 [-68388.43, -57496.30] p n/a P(B better	-141926.54 [-149473.48, -134379.60] p n/a P(B be	-298334.56 [-308948.99, -287720.13] p n/a P(B better) 1
IPPO (last) vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-62942.37 [-68388.43, -57496.30] p n/a P(B better	-141926.54 [-149473.48, -134379.60] p n/a P(B be	-298334.58 [-308949.01, -287720.14] p n/a P(B better) 1
MAPPO vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-188.49 [-226.96, -150.03] p n/a P(B better) 1.00	-503.72 [-598.54, -408.90] p n/a P(B better) 1.00	-1002.59 [-1177.43, -827.75] p n/a P(B better) 1.00
MAPPO + flip vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-188.49 [-226.96, -150.03] p n/a P(B better) 1.00	-503.71 [-598.53, -408.89] p n/a P(B better) 1.00	-1003.58 [-1178.14, -829.02] p n/a P(B better) 1.00
MAPPO (last) vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-173.79 [-220.35, -127.23] p n/a P(B better) 0.97	-501.81 [-596.35, -407.26] p n/a P(B better) 1.00	-996.18 [-1169.16, -823.21] p n/a P(B better) 1.00
Single-agent fine-tuned xN + flip vs Sin	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	+6311.41 [+3559.68, +9063.15] p n/a P(B better)	+4529.91 [+1328.61, +7731.21] p n/a P(B better)	+1497.01 [+783.74, +2210.28] p n/a P(B better) 0.11
IPPO vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-188.49 [-226.96, -150.03] p n/a P(B better) 1.00	-504.07 [-598.89, -409.26] p n/a P(B better) 1.00	-1005.11 [-1179.74, -830.49] p n/a P(B better) 1.00
IPPO + flip vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-188.49 [-226.96, -150.03] p n/a P(B better) 1.00	-504.07 [-598.89, -409.26] p n/a P(B better) 1.00	-1005.10 [-1179.72, -830.48] p n/a P(B better) 1.00
IPPO (last) vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-188.49 [-226.96, -150.03] p n/a P(B better) 1.00	-504.07 [-598.89, -409.26] p n/a P(B better) 1.00	-1005.11 [-1179.74, -830.49] p n/a P(B better) 1.00

Workload imbalance (CV): difference, lower is better

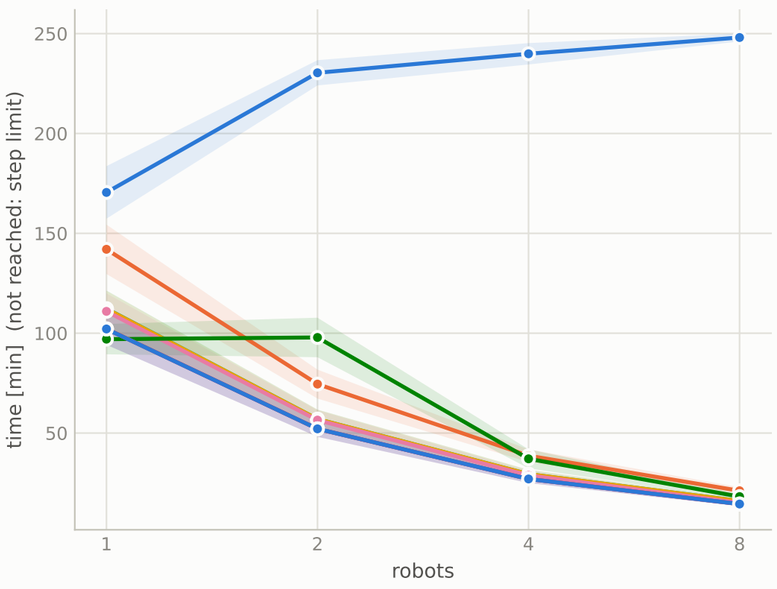
B vs A	1 robots	2 robots	4 robots	8 robots
Single-agent xN + flip vs Single-agent x	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.02 [-0.05, +0.00] p n/a P(B better) 0.66	-0.55 [-0.61, -0.49] p n/a P(B better) 1.00	-0.68 [-0.75, -0.62] p n/a P(B better) 1.00
MAPPO vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.02 [-0.05, +0.02] p n/a P(B better) 0.53	-0.57 [-0.63, -0.50] p n/a P(B better) 1.00	-0.71 [-0.77, -0.65] p n/a P(B better) 1.00
MAPPO + flip vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.02 [-0.05, +0.01] p n/a P(B better) 0.53	-0.57 [-0.63, -0.50] p n/a P(B better) 1.00	-0.71 [-0.77, -0.65] p n/a P(B better) 1.00
MAPPO (last) vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.03 [-0.05, -0.01] p n/a P(B better) 0.61	-0.55 [-0.62, -0.48] p n/a P(B better) 1.00	-0.72 [-0.78, -0.66] p n/a P(B better) 1.00
Single-agent fine-tuned xN + flip vs Sin	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.04 [-0.06, -0.01] p n/a P(B better) 0.58	-0.45 [-0.53, -0.38] p n/a P(B better) 0.97	-0.57 [-0.63, -0.51] p n/a P(B better) 1.00
IPPO vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.05 [-0.07, -0.02] p n/a P(B better) 0.79	-0.58 [-0.64, -0.51] p n/a P(B better) 1.00	-0.72 [-0.78, -0.66] p n/a P(B better) 1.00
IPPO + flip vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.05 [-0.07, -0.02] p n/a P(B better) 0.79	-0.58 [-0.64, -0.51] p n/a P(B better) 1.00	-0.72 [-0.78, -0.66] p n/a P(B better) 1.00
IPPO (last) vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.05 [-0.07, -0.02] p n/a P(B better) 0.79	-0.58 [-0.64, -0.51] p n/a P(B better) 1.00	-0.72 [-0.78, -0.66] p n/a P(B better) 1.00
MAPPO vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	+0.01 [-0.02, +0.03] p n/a P(B better) 0.50	-0.02 [-0.03, -0.00] p n/a P(B better) 0.63	-0.02 [-0.04, -0.01] p n/a P(B better) 0.63
MAPPO + flip vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.00 [-0.02, +0.02] p n/a P(B better) 0.50	-0.02 [-0.03, -0.00] p n/a P(B better) 0.63	-0.02 [-0.04, -0.01] p n/a P(B better) 0.61
MAPPO (last) vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.01 [-0.03, +0.01] p n/a P(B better) 0.58	-0.00 [-0.02, +0.02] p n/a P(B better) 0.61	-0.04 [-0.06, -0.02] p n/a P(B better) 0.76

Multi-agent coverage, test_large: method comparison

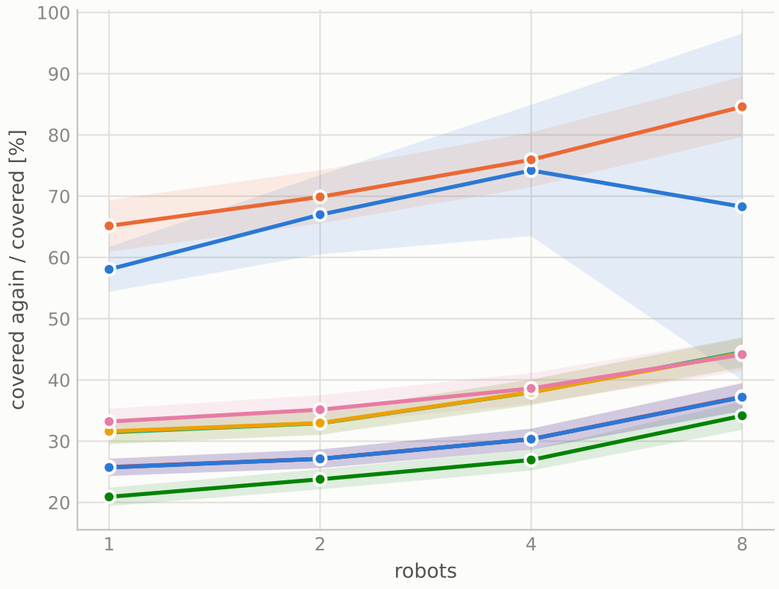
100 maps x 3 episode(s) x 1 training seed(s) per method and team size, deterministic actions, same start poses for every policy. Mean over maps of the mean over seeds, 95 % CI over seeds and maps.

Single-agent xN · Single-agent xN + flip · MAPPO · MAPPO + flip · MAPPO (last) · Single-agent fine-tuned xN + flip · IPPO · IPPO + flip · IPPO (last)

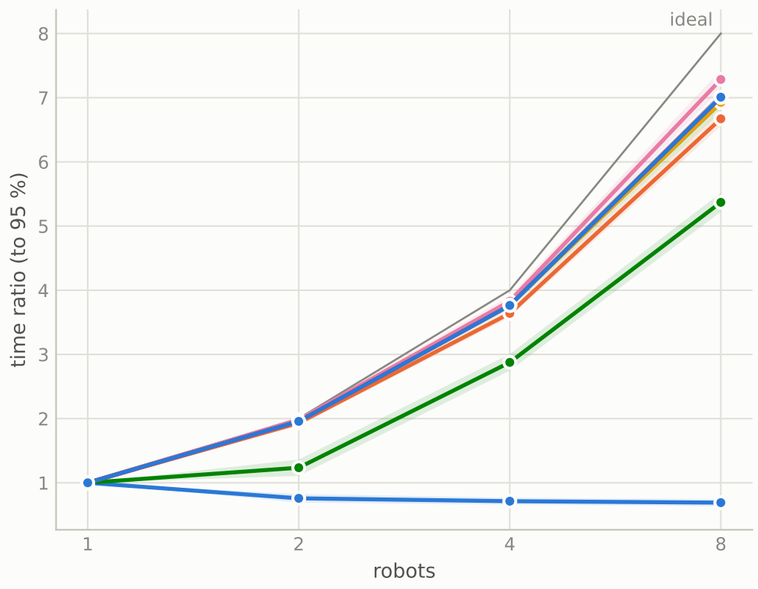
Time to 95 % coverage



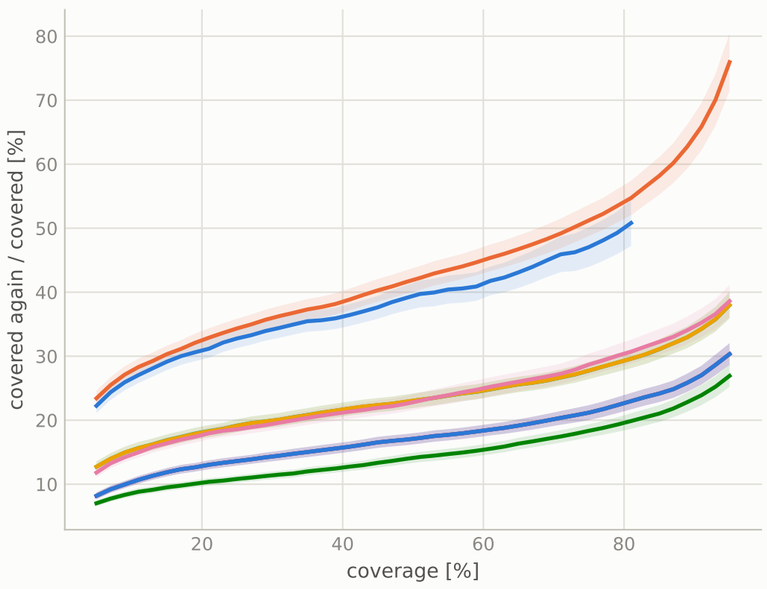
Overlap when 95 % coverage is reached



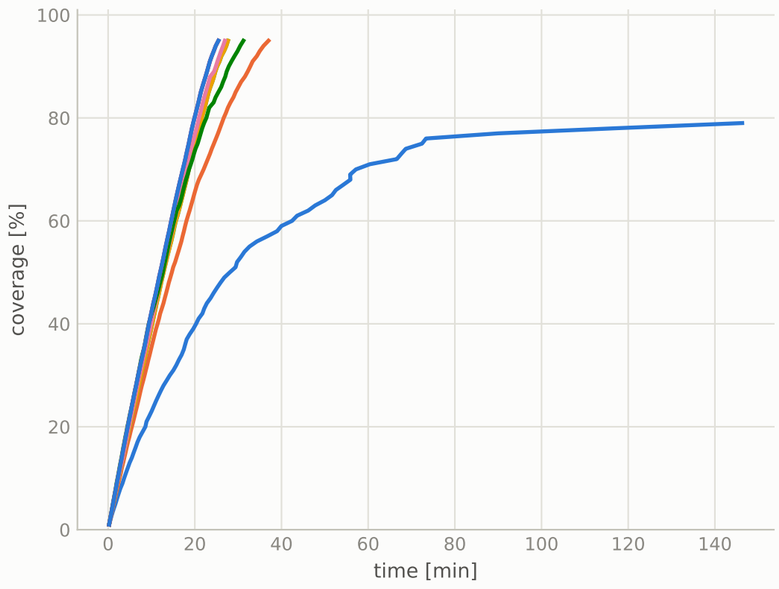
Speedup over 1 robot



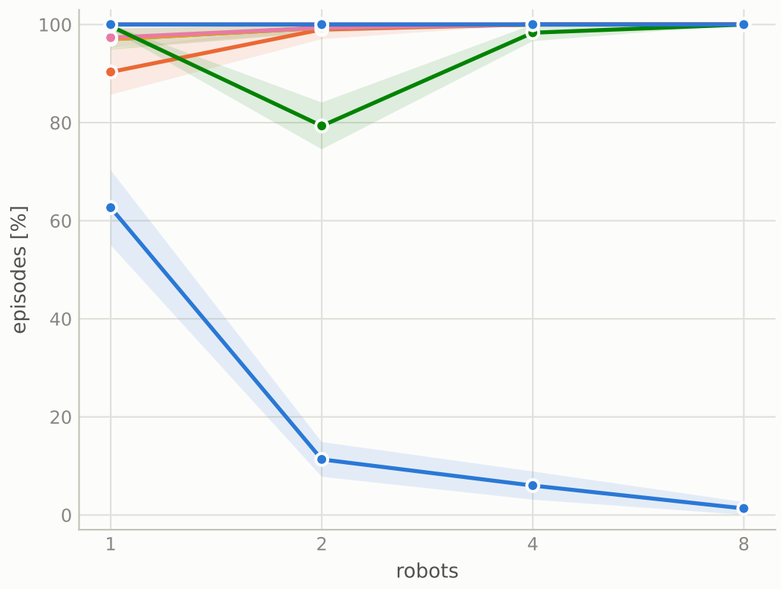
Overlap while covering, 4 robots



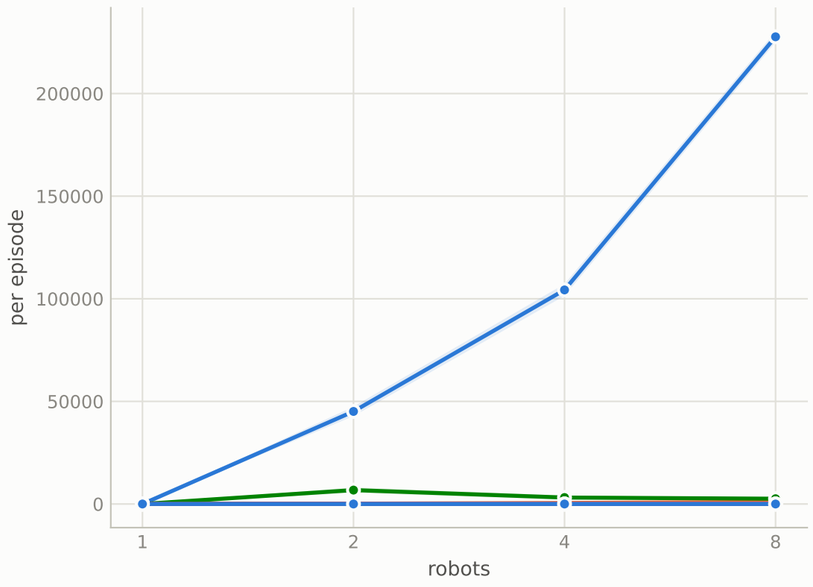
Coverage over time, 4 robots (median map)



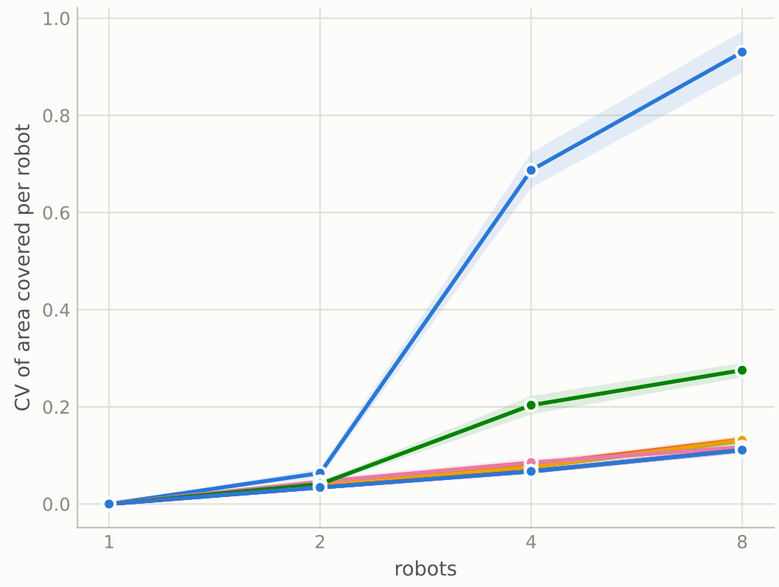
Success rate (95 % coverage)



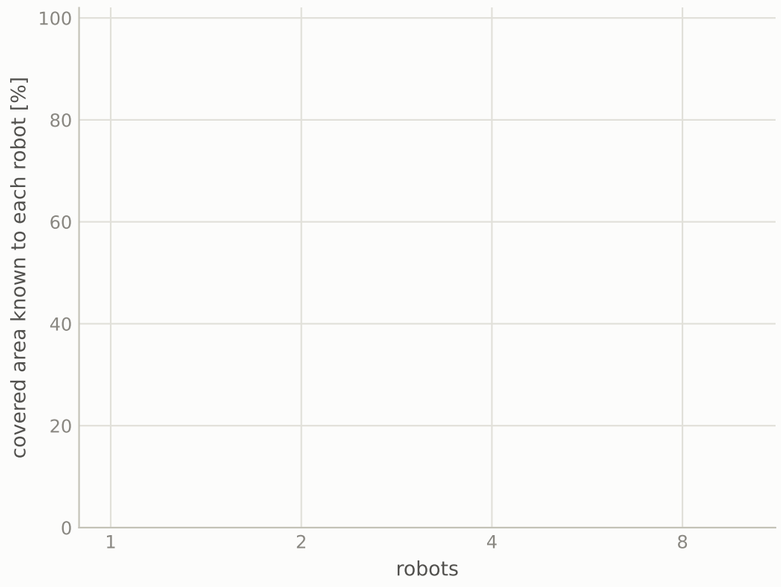
Collisions between robots



Workload imbalance



Belief recall (limited communication)



Summary, 4 robots: mean [95 % CI]

Method	Comm.	Seeds	Success (95 %)	Time to 95 %	Overlap at 95 %	Robot coll.	Imbalance	Belief recall	Connectivity
Single-agent xN	unlimited	1	6 [3, 9] %	239.9 [234.5, 245.3] mir	74.2 [63.5, 84.9] %	104310.5 [101420.1, 10	0.69 [0.65, 0.72]	100 [100, 100] %	100 [100, 100] %
Single-agent xN + flip	unlimited	1	100 [100, 100] %	38.7 [35.5, 41.8] min	75.9 [71.5, 80.4] %	513.2 [477.9, 548.5]	0.08 [0.08, 0.09]	100 [100, 100] %	100 [100, 100] %
MAPPO	unlimited	1	100 [100, 100] %	29.4 [27.3, 31.6] min	38.0 [35.9, 40.1] %	0.3 [0.2, 0.4]	0.08 [0.07, 0.08]	100 [100, 100] %	100 [100, 100] %
MAPPO + flip	unlimited	1	100 [100, 100] %	29.4 [27.3, 31.6] min	38.0 [35.9, 40.1] %	0.2 [0.1, 0.4]	0.08 [0.07, 0.08]	100 [100, 100] %	100 [100, 100] %
MAPPO (last)	unlimited	1	100 [100, 100] %	28.8 [26.6, 31.0] min	38.6 [36.1, 41.1] %	1.7 [0.5, 2.8]	0.09 [0.08, 0.09]	100 [100, 100] %	100 [100, 100] %
Single-agent fine-tuned xN + flip	unlimited	1	98 [97, 100] %	37.1 [32.3, 41.9] min	26.9 [25.2, 28.7] %	3146.0 [2095.4, 4196.7]	0.20 [0.18, 0.22]	100 [100, 100] %	100 [100, 100] %
IPPO	unlimited	1	100 [100, 100] %	27.0 [25.0, 29.0] min	30.3 [28.6, 32.0] %	0.2 [0.2, 0.3]	0.07 [0.06, 0.07]	100 [100, 100] %	100 [100, 100] %

Multi-agent coverage, test_large: statistical tests

100 maps x 3 episode(s) x 1 training seed(s) per method and team size, deterministic actions, same start poses for every policy. Mean over maps of the mean over seeds, 95 % CI over seeds and maps.

Single-agent xN · Single-agent xN + flip · MAPPO · MAPPO + flip · MAPPO (last) · Single-agent fine-tuned xN + flip · IPPO · IPPO + flip · IPPO (last)

Each cell: difference B - A [95 % CI], Holm-corrected p-value (within each metric), and P(B better): probability that B beats A on a random map, for a random seed of each (0.5 = no difference).
Bold: significant at the 5 % level after the correction. A single seed gives no p-value. Policies evaluated twice (same checkpoints) are paired by seed.

Time to the target coverage (95 %): relative difference, lower is better

B vs A	1 robots	2 robots	4 robots	8 robots
Single-agent xN + flip vs Single-agent x	-16.7 % [-21.0, -12.4] p n/a P(B better) 0.83	-67.7 % [-71.2, -64.1] p n/a P(B better) 0.99	-83.9 % [-86.1, -81.7] p n/a P(B better) 1.00	-91.5 % [-92.6, -90.4] p n/a P(B better) 1.00
MAPPO vs Single-agent xN	-34.1 % [-38.9, -29.4] p n/a P(B better) 1.00	-75.3 % [-78.2, -72.3] p n/a P(B better) 1.00	-87.7 % [-89.8, -85.6] p n/a P(B better) 1.00	-93.6 % [-94.5, -92.6] p n/a P(B better) 1.00
MAPPO + flip vs Single-agent xN	-34.2 % [-39.0, -29.4] p n/a P(B better) 1.00	-75.3 % [-78.2, -72.4] p n/a P(B better) 1.00	-87.7 % [-89.8, -85.6] p n/a P(B better) 1.00	-93.6 % [-94.5, -92.6] p n/a P(B better) 1.00
MAPPO (last) vs Single-agent xN	-34.9 % [-39.7, -30.1] p n/a P(B better) 1.00	-75.5 % [-78.4, -72.6] p n/a P(B better) 1.00	-88.0 % [-90.1, -85.9] p n/a P(B better) 1.00	-93.9 % [-94.9, -92.9] p n/a P(B better) 1.00
Single-agent fine-tuned xN + flip vs Sin	-43.1 % [-48.2, -38.0] p n/a P(B better) 0.99	-57.5 % [-62.2, -52.8] p n/a P(B better) 0.99	-84.5 % [-87.1, -82.0] p n/a P(B better) 1.00	-92.7 % [-93.8, -91.6] p n/a P(B better) 1.00
IPPO vs Single-agent xN	-40.1 % [-45.2, -35.0] p n/a P(B better) 1.00	-77.4 % [-80.1, -74.7] p n/a P(B better) 1.00	-88.7 % [-90.8, -86.6] p n/a P(B better) 1.00	-94.2 % [-95.1, -93.2] p n/a P(B better) 1.00
IPPO + flip vs Single-agent xN	-40.1 % [-45.1, -35.0] p n/a P(B better) 1.00	-77.4 % [-80.1, -74.7] p n/a P(B better) 1.00	-88.7 % [-90.8, -86.6] p n/a P(B better) 1.00	-94.2 % [-95.1, -93.2] p n/a P(B better) 1.00
IPPO (last) vs Single-agent xN	-40.1 % [-45.2, -35.0] p n/a P(B better) 1.00	-77.4 % [-80.1, -74.7] p n/a P(B better) 1.00	-88.7 % [-90.8, -86.6] p n/a P(B better) 1.00	-94.2 % [-95.1, -93.2] p n/a P(B better) 1.00
MAPPO vs Single-agent xN + flip	-20.9 % [-24.0, -17.9] p n/a P(B better) 0.98	-23.5 % [-28.7, -18.4] p n/a P(B better) 0.97	-23.8 % [-26.8, -20.9] p n/a P(B better) 1.00	-24.3 % [-27.5, -21.1] p n/a P(B better) 1.00
MAPPO + flip vs Single-agent xN + flip	-21.0 % [-24.1, -17.9] p n/a P(B better) 0.98	-23.5 % [-28.6, -18.5] p n/a P(B better) 0.98	-23.9 % [-26.8, -20.9] p n/a P(B better) 1.00	-24.4 % [-27.6, -21.2] p n/a P(B better) 1.00
MAPPO (last) vs Single-agent xN + flip	-21.9 % [-25.1, -18.7] p n/a P(B better) 0.99	-24.2 % [-28.0, -20.4] p n/a P(B better) 0.99	-25.5 % [-28.4, -22.6] p n/a P(B better) 1.00	-28.5 % [-31.8, -25.3] p n/a P(B better) 1.00
Single-agent fine-tuned xN + flip vs Sin	-31.7 % [-35.6, -27.8] p n/a P(B better) 0.99	+31.3 % [+16.9, +45.8] p n/a P(B better) 0.48	-4.0 % [-14.3, +6.2] p n/a P(B better) 0.84	-13.9 % [-17.0, -10.7] p n/a P(B better) 0.89
IPPO vs Single-agent xN + flip	-28.1 % [-31.7, -24.5] p n/a P(B better) 1.00	-30.1 % [-35.3, -24.8] p n/a P(B better) 1.00	-30.1 % [-33.5, -26.8] p n/a P(B better) 1.00	-31.5 % [-35.0, -27.9] p n/a P(B better) 1.00
IPPO + flip vs Single-agent xN + flip	-28.0 % [-31.6, -24.4] p n/a P(B better) 1.00	-30.1 % [-35.3, -24.8] p n/a P(B better) 1.00	-30.1 % [-33.5, -26.8] p n/a P(B better) 1.00	-31.4 % [-35.0, -27.9] p n/a P(B better) 1.00
IPPO (last) vs Single-agent xN + flip	-28.1 % [-31.7, -24.5] p n/a P(B better) 1.00	-30.1 % [-35.3, -24.8] p n/a P(B better) 1.00	-30.1 % [-33.5, -26.8] p n/a P(B better) 1.00	-31.5 % [-35.0, -27.9] p n/a P(B better) 1.00

Overlap when the target coverage is reached (95 %): percentage points, lower is better

B vs A	1 robots	2 robots	4 robots	8 robots
Single-agent xN + flip vs Single-agent x	+7.1 pp [+5.2, +9.0] p n/a P(B better) 0.46	+2.9 pp [-1.8, +7.6] p n/a P(B better) 0.26	+1.7 pp [-5.9, +9.3] p n/a P(B better) 0.38	+16.3 pp [-11.9, +44.5] p n/a P(B better) 0.00
MAPPO vs Single-agent xN	-26.6 pp [-29.1, -24.1] p n/a P(B better) 1.00	-34.1 pp [-38.8, -29.3] p n/a P(B better) 1.00	-36.2 pp [-45.9, -26.5] p n/a P(B better) 0.94	-23.7 pp [-36.1, -11.4] p n/a P(B better) 1.00
MAPPO + flip vs Single-agent xN	-26.4 pp [-28.9, -24.0] p n/a P(B better) 1.00	-34.0 pp [-38.7, -29.3] p n/a P(B better) 1.00	-36.2 pp [-45.9, -26.5] p n/a P(B better) 0.94	-23.9 pp [-37.1, -10.7] p n/a P(B better) 1.00
MAPPO (last) vs Single-agent xN	-24.8 pp [-27.3, -22.4] p n/a P(B better) 1.00	-31.8 pp [-36.9, -26.8] p n/a P(B better) 1.00	-35.6 pp [-43.2, -28.0] p n/a P(B better) 1.00	-24.1 pp [-35.6, -12.6] p n/a P(B better) 1.00
Single-agent fine-tuned xN + flip vs Sin	-37.1 pp [-39.6, -34.7] p n/a P(B better) 1.00	-43.2 pp [-48.6, -37.8] p n/a P(B better) 1.00	-47.3 pp [-56.1, -38.4] p n/a P(B better) 1.00	-34.1 pp [-40.0, -28.3] p n/a P(B better) 1.00
IPPO vs Single-agent xN	-32.3 pp [-35.3, -29.4] p n/a P(B better) 1.00	-39.9 pp [-45.1, -34.7] p n/a P(B better) 1.00	-43.9 pp [-52.5, -35.2] p n/a P(B better) 1.00	-31.1 pp [-35.0, -27.1] p n/a P(B better) 1.00
IPPO + flip vs Single-agent xN	-32.3 pp [-35.2, -29.3] p n/a P(B better) 1.00	-39.9 pp [-45.0, -34.7] p n/a P(B better) 1.00	-43.9 pp [-52.5, -35.2] p n/a P(B better) 1.00	-31.0 pp [-35.0, -27.1] p n/a P(B better) 1.00
IPPO (last) vs Single-agent xN	-32.3 pp [-35.3, -29.4] p n/a P(B better) 1.00	-39.9 pp [-45.1, -34.7] p n/a P(B better) 1.00	-43.9 pp [-52.5, -35.2] p n/a P(B better) 1.00	-31.1 pp [-35.0, -27.1] p n/a P(B better) 1.00
MAPPO vs Single-agent xN + flip	-33.7 pp [-36.6, -30.9] p n/a P(B better) 1.00	-37.0 pp [-40.2, -33.7] p n/a P(B better) 1.00	-38.0 pp [-41.3, -34.6] p n/a P(B better) 1.00	-40.1 pp [-43.7, -36.4] p n/a P(B better) 1.00
MAPPO + flip vs Single-agent xN + flip	-33.5 pp [-36.4, -30.7] p n/a P(B better) 1.00	-36.9 pp [-40.2, -33.7] p n/a P(B better) 1.00	-37.9 pp [-41.3, -34.6] p n/a P(B better) 1.00	-40.3 pp [-44.0, -36.5] p n/a P(B better) 1.00
MAPPO (last) vs Single-agent xN + flip	-31.9 pp [-34.8, -29.1] p n/a P(B better) 1.00	-34.7 pp [-37.8, -31.7] p n/a P(B better) 1.00	-37.3 pp [-40.3, -34.3] p n/a P(B better) 1.00	-40.5 pp [-43.9, -37.0] p n/a P(B better) 1.00
Single-agent fine-tuned xN + flip vs Sin	-44.2 pp [-47.3, -41.1] p n/a P(B better) 1.00	-46.1 pp [-49.5, -42.7] p n/a P(B better) 1.00	-49.0 pp [-52.5, -45.6] p n/a P(B better) 1.00	-50.4 pp [-54.1, -46.8] p n/a P(B better) 1.00
IPPO vs Single-agent xN + flip	-39.4 pp [-42.7, -36.2] p n/a P(B better) 1.00	-42.8 pp [-46.2, -39.4] p n/a P(B better) 1.00	-45.6 pp [-49.1, -42.1] p n/a P(B better) 1.00	-47.4 pp [-51.1, -43.7] p n/a P(B better) 1.00
IPPO + flip vs Single-agent xN + flip	-39.4 pp [-42.6, -36.1] p n/a P(B better) 1.00	-42.8 pp [-46.2, -39.4] p n/a P(B better) 1.00	-45.6 pp [-49.1, -42.1] p n/a P(B better) 1.00	-47.3 pp [-51.0, -43.7] p n/a P(B better) 1.00
IPPO (last) vs Single-agent xN + flip	-39.4 pp [-42.7, -36.2] p n/a P(B better) 1.00	-42.8 pp [-46.2, -39.4] p n/a P(B better) 1.00	-45.6 pp [-49.1, -42.1] p n/a P(B better) 1.00	-47.4 pp [-51.1, -43.7] p n/a P(B better) 1.00

Success rate (95 %): percentage points, higher is better

B vs A	1 robots	2 robots	4 robots	8 robots
Single-agent xN + flip vs Single-agent x	+27.7 pp [+21.1, +34.2] p n/a P(B better) 0.76	+87.7 pp [+83.7, +91.6] p n/a P(B better) 0.99	+94.0 pp [+91.1, +96.9] p n/a P(B better) 1.00	+98.7 pp [+97.4, +100.0] p n/a P(B better) 1.00
MAPPO vs Single-agent xN	+34.3 pp [+26.7, +42.0] p n/a P(B better) 0.77	+88.0 pp [+84.3, +91.7] p n/a P(B better) 1.00	+94.0 pp [+91.1, +96.9] p n/a P(B better) 1.00	+98.7 pp [+97.4, +100.0] p n/a P(B better) 1.00
MAPPO + flip vs Single-agent xN	+34.3 pp [+26.7, +42.0] p n/a P(B better) 0.77	+88.0 pp [+84.3, +91.7] p n/a P(B better) 1.00	+94.0 pp [+91.1, +96.9] p n/a P(B better) 1.00	+98.7 pp [+97.4, +100.0] p n/a P(B better) 1.00
MAPPO (last) vs Single-agent xN	+34.7 pp [+27.2, +42.2] p n/a P(B better) 0.78	+88.0 pp [+84.4, +91.6] p n/a P(B better) 1.00	+94.0 pp [+91.1, +96.9] p n/a P(B better) 1.00	+98.7 pp [+97.4, +100.0] p n/a P(B better) 1.00
Single-agent fine-tuned xN + flip vs Sin	+37.0 pp [+29.3, +44.7] p n/a P(B better) 0.79	+68.0 pp [+62.2, +73.8] p n/a P(B better) 0.98	+92.3 pp [+89.1, +95.6] p n/a P(B better) 1.00	+98.7 pp [+97.4, +100.0] p n/a P(B better) 1.00
IPPO vs Single-agent xN	+37.3 pp [+29.7, +45.0] p n/a P(B better) 0.80	+88.7 pp [+85.1, +92.2] p n/a P(B better) 1.00	+94.0 pp [+91.1, +96.9] p n/a P(B better) 1.00	+98.7 pp [+97.4, +100.0] p n/a P(B better) 1.00
IPPO + flip vs Single-agent xN	+37.3 pp [+29.7, +45.0] p n/a P(B better) 0.80	+88.7 pp [+85.1, +92.2] p n/a P(B better) 1.00	+94.0 pp [+91.1, +96.9] p n/a P(B better) 1.00	+98.7 pp [+97.4, +100.0] p n/a P(B better) 1.00
IPPO (last) vs Single-agent xN	+37.3 pp [+29.7, +45.0] p n/a P(B better) 0.80	+88.7 pp [+85.1, +92.2] p n/a P(B better) 1.00	+94.0 pp [+91.1, +96.9] p n/a P(B better) 1.00	+98.7 pp [+97.4, +100.0] p n/a P(B better) 1.00
MAPPO vs Single-agent xN + flip	+6.7 pp [+2.2, +11.2] p n/a P(B better) 0.56	+0.3 pp [-1.9, +2.5] p n/a P(B better) 0.49	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50
MAPPO + flip vs Single-agent xN + flip	+6.7 pp [+2.2, +11.2] p n/a P(B better) 0.56	+0.3 pp [-1.9, +2.5] p n/a P(B better) 0.49	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50
MAPPO (last) vs Single-agent xN + flip	+7.0 pp [+2.3, +11.7] p n/a P(B better) 0.56	+0.3 pp [-1.2, +1.8] p n/a P(B better) 0.50	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50
Single-agent fine-tuned xN + flip vs Sin	+9.3 pp [+4.6, +14.0] p n/a P(B better) 0.58	-19.7 pp [-24.7, -14.6] p n/a P(B better) 0.27	-1.7 pp [-3.4, +0.1] p n/a P(B better) 0.48	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50
IPPO vs Single-agent xN + flip	+9.7 pp [+5.0, +14.3] p n/a P(B better) 0.59	+1.0 pp [-1.0, +3.0] p n/a P(B better) 0.51	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50
IPPO + flip vs Single-agent xN + flip	+9.7 pp [+5.0, +14.3] p n/a P(B better) 0.59	+1.0 pp [-1.0, +3.0] p n/a P(B better) 0.51	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50
IPPO (last) vs Single-agent xN + flip	+9.7 pp [+5.0, +14.3] p n/a P(B better) 0.59	+1.0 pp [-1.0, +3.0] p n/a P(B better) 0.51	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50

Collisions between robots per episode: difference, lower is better

B vs A	1 robots	2 robots	4 robots	8 robots
Single-agent xN + flip vs Single-agent x	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-44923.98 [-47255.24, -42592.72] p n/a P(B better	-103797.38 [-106686.06, -100908.70] p n/a P(B be	-226600.82 [-229248.44, -223953.20] p n/a P(B better) 1
MAPPO vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-45099.40 [-47433.77, -42765.03] p n/a P(B better	-104310.25 [-107200.71, -101419.80] p n/a P(B be	-227638.02 [-230277.38, -224998.66] p n/a P(B better) 1
MAPPO + flip vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-45101.46 [-47436.23, -42766.69] p n/a P(B better	-104310.29 [-107200.74, -101419.85] p n/a P(B be	-227640.29 [-230279.90, -225000.68] p n/a P(B better) 1
MAPPO (last) vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-45097.91 [-47433.91, -42761.90] p n/a P(B better	-104308.85 [-107199.43, -101418.27] p n/a P(B be	-227621.88 [-230261.59, -224982.17] p n/a P(B better) 1
Single-agent fine-tuned xN + flip vs Sin	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-38325.12 [-40832.29, -35817.94] p n/a P(B better	-101164.50 [-104118.58, -98210.43] p n/a P(B bett	-225065.72 [-227849.43, -222282.01] p n/a P(B better) 1
IPPO vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-45102.94 [-47437.99, -42767.88] p n/a P(B better	-104310.31 [-107200.75, -101419.87] p n/a P(B be	-227643.64 [-230283.25, -225004.03] p n/a P(B better) 1
IPPO + flip vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-45102.94 [-47437.99, -42767.88] p n/a P(B better	-104310.30 [-107200.74, -101419.86] p n/a P(B be	-227643.67 [-230283.28, -225004.06] p n/a P(B better) 1
IPPO (last) vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-45102.94 [-47437.99, -42767.88] p n/a P(B better	-104310.31 [-107200.75, -101419.87] p n/a P(B be	-227643.64 [-230283.25, -225004.03] p n/a P(B better) 1
MAPPO vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-175.42 [-195.81, -155.03] p n/a P(B better) 0.99	-512.87 [-548.16, -477.59] p n/a P(B better) 1.00	-1037.20 [-1098.19, -976.21] p n/a P(B better) 1.00
MAPPO + flip vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-177.48 [-196.32, -158.64] p n/a P(B better) 0.99	-512.91 [-548.20, -477.62] p n/a P(B better) 1.00	-1039.48 [-1100.11, -978.84] p n/a P(B better) 1.00
MAPPO (last) vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-173.92 [-195.53, -152.32] p n/a P(B better) 0.99	-511.47 [-546.81, -476.13] p n/a P(B better) 1.00	-1021.06 [-1083.63, -958.50] p n/a P(B better) 1.00
Single-agent fine-tuned xN + flip vs Sin	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	+6598.87 [+5260.44, +7937.30] p n/a P(B better)	+2632.88 [+1586.12, +3679.63] p n/a P(B better)	+1535.10 [+1273.18, +1797.02] p n/a P(B better) 0.00
IPPO vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-178.95 [-197.19, -160.72] p n/a P(B better) 1.00	-512.93 [-548.23, -477.63] p n/a P(B better) 1.00	-1042.82 [-1103.56, -982.09] p n/a P(B better) 1.00
IPPO + flip vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-178.95 [-197.19, -160.72] p n/a P(B better) 1.00	-512.92 [-548.22, -477.63] p n/a P(B better) 1.00	-1042.85 [-1103.59, -982.12] p n/a P(B better) 1.00
IPPO (last) vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-178.95 [-197.19, -160.72] p n/a P(B better) 1.00	-512.93 [-548.23, -477.63] p n/a P(B better) 1.00	-1042.82 [-1103.56, -982.09] p n/a P(B better) 1.00

Workload imbalance (CV): difference, lower is better

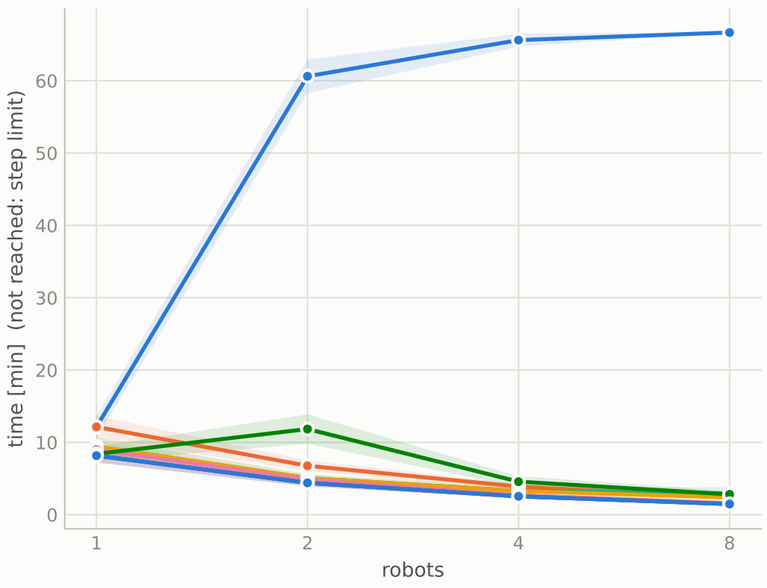
B vs A	1 robots	2 robots	4 robots	8 robots
Single-agent xN + flip vs Single-agent x	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.02 [-0.03, -0.01] p n/a P(B better) 0.61	-0.61 [-0.64, -0.57] p n/a P(B better) 1.00	-0.80 [-0.84, -0.76] p n/a P(B better) 1.00
MAPPO vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.03 [-0.03, -0.02] p n/a P(B better) 0.75	-0.61 [-0.65, -0.58] p n/a P(B better) 1.00	-0.80 [-0.84, -0.76] p n/a P(B better) 1.00
MAPPO + flip vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.02 [-0.03, -0.02] p n/a P(B better) 0.75	-0.61 [-0.65, -0.58] p n/a P(B better) 1.00	-0.80 [-0.84, -0.76] p n/a P(B better) 1.00
MAPPO (last) vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.02 [-0.03, -0.01] p n/a P(B better) 0.70	-0.60 [-0.64, -0.57] p n/a P(B better) 1.00	-0.81 [-0.86, -0.77] p n/a P(B better) 1.00
Single-agent fine-tuned xN + flip vs Sin	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.02 [-0.03, -0.01] p n/a P(B better) 0.69	-0.48 [-0.52, -0.45] p n/a P(B better) 1.00	-0.65 [-0.70, -0.61] p n/a P(B better) 1.00
IPPO vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.03 [-0.04, -0.02] p n/a P(B better) 0.81	-0.62 [-0.66, -0.58] p n/a P(B better) 1.00	-0.82 [-0.86, -0.78] p n/a P(B better) 1.00
IPPO + flip vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.03 [-0.04, -0.02] p n/a P(B better) 0.81	-0.62 [-0.66, -0.58] p n/a P(B better) 1.00	-0.82 [-0.86, -0.78] p n/a P(B better) 1.00
IPPO (last) vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.03 [-0.04, -0.02] p n/a P(B better) 0.81	-0.62 [-0.66, -0.58] p n/a P(B better) 1.00	-0.82 [-0.86, -0.78] p n/a P(B better) 1.00
MAPPO vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.01 [-0.01, -0.00] p n/a P(B better) 0.59	-0.01 [-0.01, -0.00] p n/a P(B better) 0.59	-0.00 [-0.01, +0.01] p n/a P(B better) 0.60
MAPPO + flip vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.01 [-0.01, -0.00] p n/a P(B better) 0.57	-0.01 [-0.01, +0.00] p n/a P(B better) 0.58	-0.00 [-0.01, +0.01] p n/a P(B better) 0.59
MAPPO (last) vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	+0.00 [-0.01, +0.01] p n/a P(B better) 0.54	+0.00 [-0.00, +0.01] p n/a P(B better) 0.47	-0.02 [-0.02, -0.01] p n/a P(B better) 0.66

Multi-agent coverage, test_small: method comparison

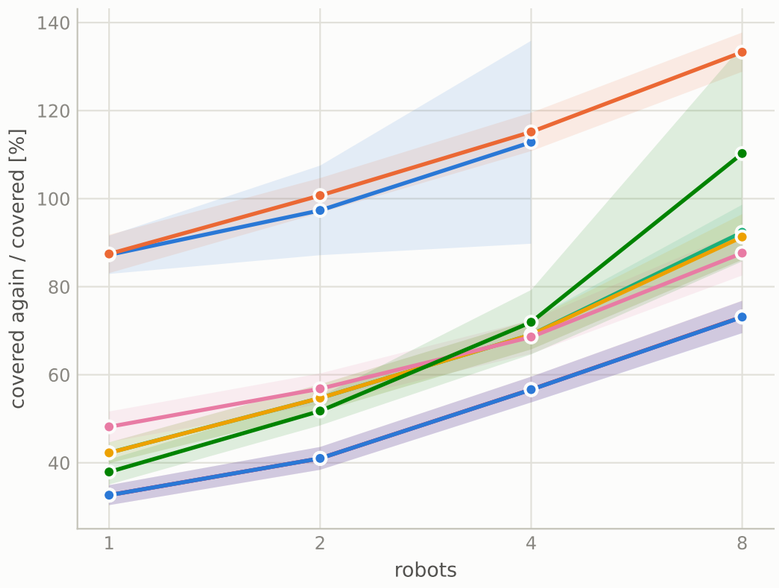
100 maps x 3 episode(s) x 1 training seed(s) per method and team size, deterministic actions, same start poses for every policy. Mean over maps of the mean over seeds, 95 % CI over seeds and maps.

Single-agent xN · Single-agent xN + flip · MAPPO · MAPPO + flip · MAPPO (last) · Single-agent fine-tuned xN + flip · IPPO · IPPO + flip · IPPO (last)

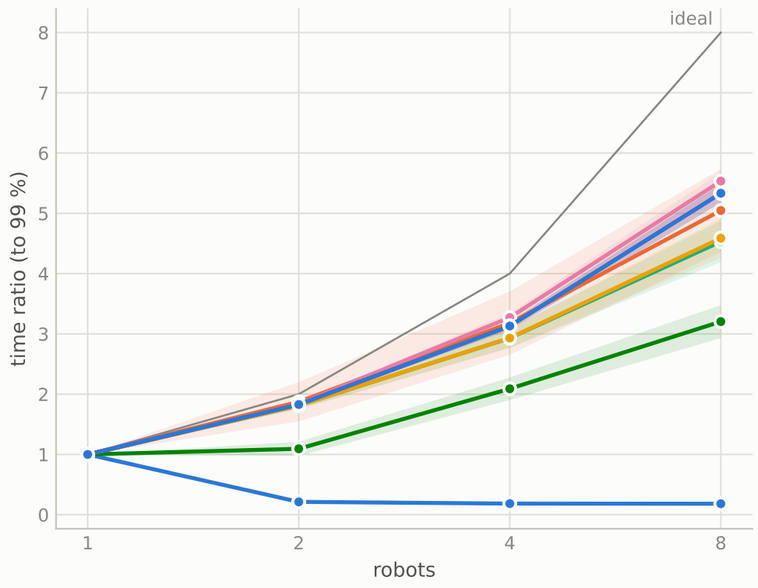
Time to 99 % coverage



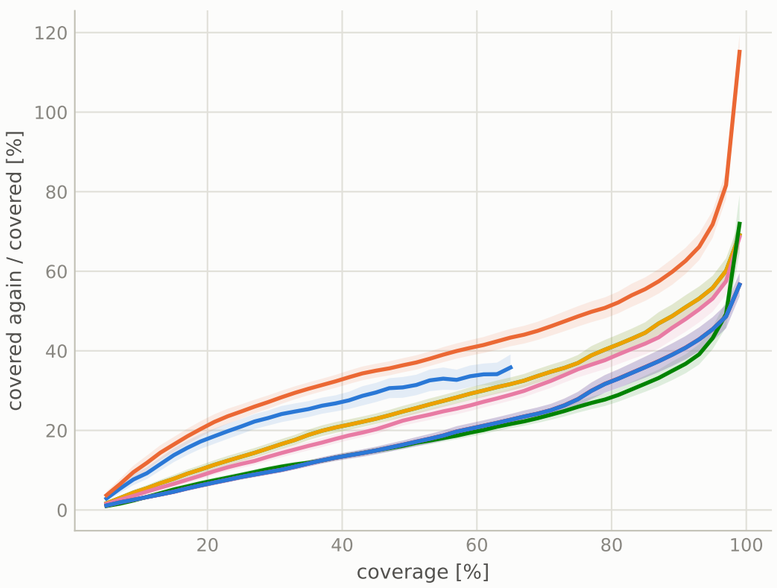
Overlap when 99 % coverage is reached



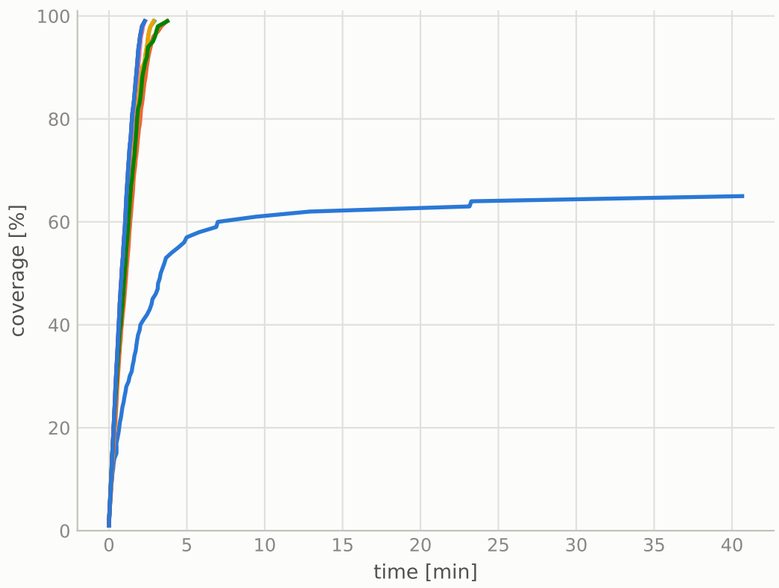
Speedup over 1 robot



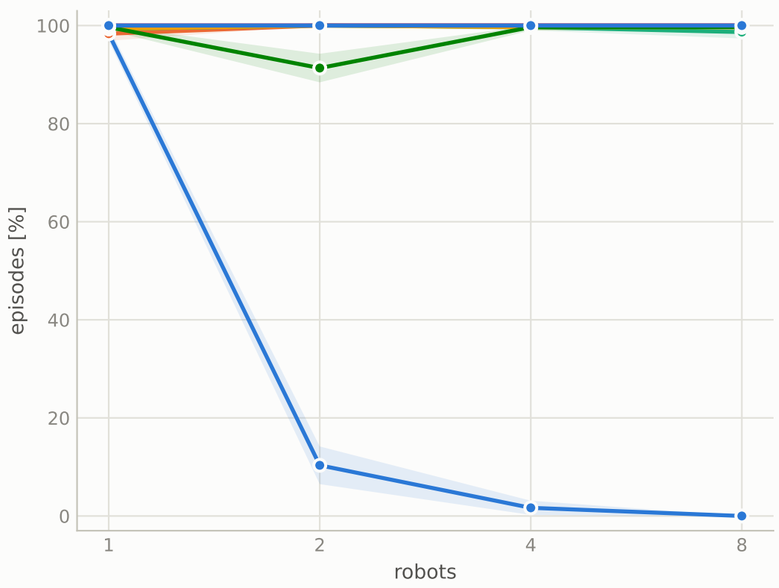
Overlap while covering, 4 robots



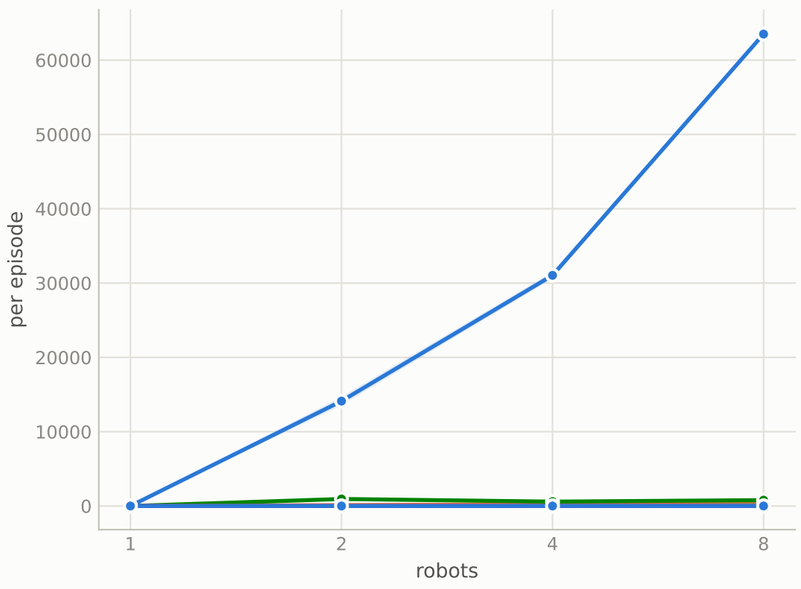
Coverage over time, 4 robots (median map)



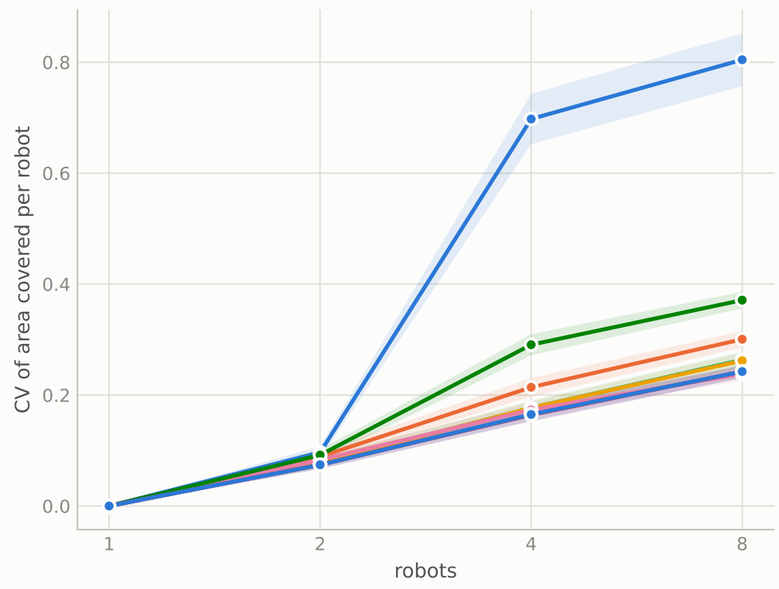
Success rate (99 % coverage)



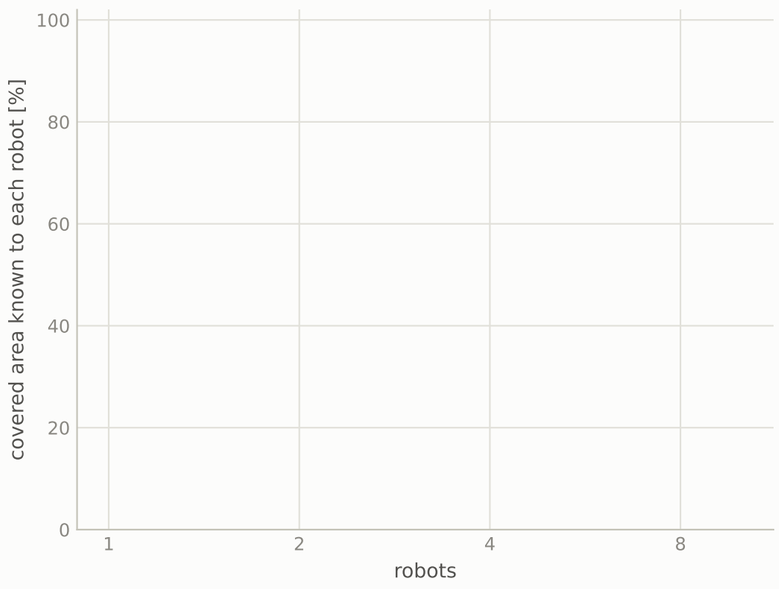
Collisions between robots



Workload imbalance



Belief recall (limited communication)



Summary, 4 robots: mean [95 % CI]

Method	Comm.	Seeds	Success (99 %)	Time to 99 %	Overlap at 99 %	Robot coll.	Imbalance	Belief recall	Connectivity
Single-agent xN	unlimited	1	2 [0, 3] %	65.6 [64.7, 66.5] min	112.8 [89.8, 135.9] %	31039.5 [30592.1, 3148	0.70 [0.65, 0.74]	100 [100, 100] %	100 [100, 100] %
Single-agent xN + flip	unlimited	1	100 [100, 100] %	3.9 [3.5, 4.3] min	115.2 [110.8, 119.5] %	225.1 [206.9, 243.4]	0.21 [0.20, 0.23]	100 [100, 100] %	100 [100, 100] %
MAPPO	unlimited	1	100 [99, 100] %	3.3 [2.8, 3.8] min	69.1 [65.8, 72.4] %	2.0 [0.0, 4.6]	0.18 [0.16, 0.19]	100 [100, 100] %	100 [100, 100] %
MAPPO + flip	unlimited	1	100 [99, 100] %	3.3 [2.8, 3.8] min	69.1 [65.7, 72.4] %	1.5 [0.0, 3.1]	0.18 [0.16, 0.19]	100 [100, 100] %	100 [100, 100] %
MAPPO (last)	unlimited	1	100 [100, 100] %	2.7 [2.4, 3.0] min	68.6 [64.8, 72.4] %	0.4 [0.2, 0.6]	0.17 [0.16, 0.19]	100 [100, 100] %	100 [100, 100] %
Single-agent fine-tuned xN + flip	unlimited	1	100 [99, 100] %	4.6 [3.8, 5.3] min	71.9 [64.6, 79.3] %	586.9 [395.9, 777.8]	0.29 [0.27, 0.31]	100 [100, 100] %	100 [100, 100] %
IPPO	unlimited	1	100 [100, 100] %	2.5 [2.3, 2.8] min	56.6 [53.7, 59.6] %	0.1 [0.1, 0.2]	0.16 [0.15, 0.18]	100 [100, 100] %	100 [100, 100] %

Multi-agent coverage, test_small: statistical tests

100 maps x 3 episode(s) x 1 training seed(s) per method and team size, deterministic actions, same start poses for every policy. Mean over maps of the mean over seeds, 95 % CI over seeds and maps.

Single-agent xN · Single-agent xN + flip · MAPPO · MAPPO + flip · MAPPO (last) · Single-agent fine-tuned xN + flip · IPPO · IPPO + flip · IPPO (last)

Each cell: difference B - A [95 % CI], Holm-corrected p-value (within each metric), and P(B better): probability that B beats A on a random map, for a random seed of each (0.5 = no difference).
Bold: significant at the 5 % level after the correction. A single seed gives no p-value. Policies evaluated twice (same checkpoints) are paired by seed.

Time to the target coverage (99 %): relative difference, lower is better

B vs A	1 robots	2 robots	4 robots	8 robots
Single-agent xN + flip vs Single-agent x	+0.1 % [-0.1, +0.3] p n/a P(B better) 0.48	-88.8 % [-92.7, -85.0] p n/a P(B better) 1.00	-94.1 % [-95.5, -92.6] p n/a P(B better) 1.00	-96.4 % [-96.8, -96.1] p n/a P(B better) 1.00
MAPPO vs Single-agent xN	-22.3 % [-30.4, -14.3] p n/a P(B better) 0.94	-91.6 % [-95.5, -87.8] p n/a P(B better) 1.00	-95.0 % [-96.5, -93.5] p n/a P(B better) 1.00	-95.6 % [-96.9, -94.3] p n/a P(B better) 1.00
MAPPO + flip vs Single-agent xN	-22.3 % [-30.4, -14.3] p n/a P(B better) 0.94	-91.6 % [-95.5, -87.8] p n/a P(B better) 1.00	-95.0 % [-96.5, -93.5] p n/a P(B better) 1.00	-96.4 % [-97.2, -95.6] p n/a P(B better) 1.00
MAPPO (last) vs Single-agent xN	-25.5 % [-32.8, -18.3] p n/a P(B better) 0.96	-92.0 % [-95.8, -88.2] p n/a P(B better) 1.00	-95.9 % [-97.3, -94.5] p n/a P(B better) 1.00	-97.6 % [-97.9, -97.4] p n/a P(B better) 1.00
Single-agent fine-tuned xN + flip vs Sin	-30.7 % [-37.5, -23.9] p n/a P(B better) 0.99	-80.5 % [-85.3, -75.6] p n/a P(B better) 0.99	-93.0 % [-94.7, -91.3] p n/a P(B better) 1.00	-95.8 % [-96.5, -95.1] p n/a P(B better) 1.00
IPPO vs Single-agent xN	-32.7 % [-40.2, -25.2] p n/a P(B better) 1.00	-92.7 % [-96.5, -88.9] p n/a P(B better) 1.00	-96.1 % [-97.5, -94.7] p n/a P(B better) 1.00	-97.8 % [-98.0, -97.6] p n/a P(B better) 1.00
IPPO + flip vs Single-agent xN	-32.7 % [-40.2, -25.2] p n/a P(B better) 1.00	-92.7 % [-96.5, -88.9] p n/a P(B better) 1.00	-96.1 % [-97.5, -94.7] p n/a P(B better) 1.00	-97.8 % [-98.0, -97.6] p n/a P(B better) 1.00
IPPO (last) vs Single-agent xN	-32.7 % [-40.2, -25.2] p n/a P(B better) 1.00	-92.7 % [-96.5, -88.9] p n/a P(B better) 1.00	-96.1 % [-97.5, -94.7] p n/a P(B better) 1.00	-97.8 % [-98.0, -97.6] p n/a P(B better) 1.00
MAPPO vs Single-agent xN + flip	-22.4 % [-30.4, -14.4] p n/a P(B better) 0.94	-25.0 % [-28.8, -21.3] p n/a P(B better) 0.98	-15.4 % [-26.7, -4.1] p n/a P(B better) 0.89	+23.9 % [-14.1, +61.9] p n/a P(B better) 0.82
MAPPO + flip vs Single-agent xN + flip	-22.4 % [-30.4, -14.4] p n/a P(B better) 0.94	-25.0 % [-28.8, -21.3] p n/a P(B better) 0.98	-15.6 % [-26.8, -4.3] p n/a P(B better) 0.89	+0.9 % [-21.9, +23.8] p n/a P(B better) 0.83
MAPPO (last) vs Single-agent xN + flip	-25.6 % [-32.8, -18.4] p n/a P(B better) 0.96	-28.3 % [-32.2, -24.4] p n/a P(B better) 1.00	-30.9 % [-34.4, -27.4] p n/a P(B better) 1.00	-33.5 % [-37.1, -30.0] p n/a P(B better) 0.99
Single-agent fine-tuned xN + flip vs Sin	-30.8 % [-37.6, -23.9] p n/a P(B better) 0.99	+75.1 % [+47.3, +102.9] p n/a P(B better) 0.53	+17.6 % [-0.9, +36.1] p n/a P(B better) 0.54	+18.3 % [-1.7, +38.4] p n/a P(B better) 0.42
IPPO vs Single-agent xN + flip	-32.7 % [-40.2, -25.2] p n/a P(B better) 1.00	-34.8 % [-39.3, -30.3] p n/a P(B better) 1.00	-34.5 % [-38.3, -30.7] p n/a P(B better) 1.00	-37.9 % [-41.7, -34.0] p n/a P(B better) 0.99
IPPO + flip vs Single-agent xN + flip	-32.7 % [-40.2, -25.2] p n/a P(B better) 1.00	-34.8 % [-39.3, -30.3] p n/a P(B better) 1.00	-34.6 % [-38.4, -30.8] p n/a P(B better) 1.00	-37.8 % [-41.7, -34.0] p n/a P(B better) 0.99
IPPO (last) vs Single-agent xN + flip	-32.7 % [-40.2, -25.2] p n/a P(B better) 1.00	-34.8 % [-39.3, -30.3] p n/a P(B better) 1.00	-34.5 % [-38.3, -30.7] p n/a P(B better) 1.00	-37.9 % [-41.7, -34.0] p n/a P(B better) 0.99

Overlap when the target coverage is reached (99 %): percentage points, lower is better

B vs A	1 robots	2 robots	4 robots	8 robots
Single-agent xN + flip vs Single-agent x	+0.2 pp [-0.1, +0.5] p n/a P(B better) 0.49	+3.4 pp [-3.6, +10.3] p n/a P(B better) 0.42	+2.3 pp [-24.5, +29.2] p n/a P(B better) 0.40	-
MAPPO vs Single-agent xN	-45.0 pp [-48.0, -41.9] p n/a P(B better) 1.00	-42.6 pp [-51.9, -33.4] p n/a P(B better) 1.00	-43.7 pp [-71.7, -15.7] p n/a P(B better) 1.00	-
MAPPO + flip vs Single-agent xN	-45.0 pp [-48.1, -41.9] p n/a P(B better) 1.00	-42.6 pp [-51.9, -33.4] p n/a P(B better) 1.00	-43.7 pp [-71.7, -15.7] p n/a P(B better) 1.00	-
MAPPO (last) vs Single-agent xN	-39.1 pp [-42.5, -35.6] p n/a P(B better) 0.98	-40.5 pp [-47.3, -33.7] p n/a P(B better) 1.00	-44.2 pp [-76.0, -12.5] p n/a P(B better) 1.00	-
Single-agent fine-tuned xN + flip vs Sin	-49.3 pp [-52.2, -46.4] p n/a P(B better) 1.00	-45.6 pp [-53.8, -37.4] p n/a P(B better) 1.00	-40.9 pp [-64.0, -17.8] p n/a P(B better) 1.00	-
IPPO vs Single-agent xN	-54.6 pp [-57.8, -51.4] p n/a P(B better) 1.00	-56.3 pp [-65.1, -47.5] p n/a P(B better) 1.00	-56.2 pp [-70.3, -42.1] p n/a P(B better) 1.00	-
IPPO + flip vs Single-agent xN	-54.6 pp [-57.8, -51.4] p n/a P(B better) 1.00	-56.3 pp [-65.1, -47.5] p n/a P(B better) 1.00	-56.2 pp [-70.2, -42.1] p n/a P(B better) 1.00	-
IPPO (last) vs Single-agent xN	-54.6 pp [-57.8, -51.4] p n/a P(B better) 1.00	-56.3 pp [-65.1, -47.5] p n/a P(B better) 1.00	-56.2 pp [-70.3, -42.1] p n/a P(B better) 1.00	-
MAPPO vs Single-agent xN + flip	-45.2 pp [-48.3, -42.1] p n/a P(B better) 1.00	-46.0 pp [-49.3, -42.7] p n/a P(B better) 0.99	-46.1 pp [-49.5, -42.6] p n/a P(B better) 1.00	-40.9 pp [-47.0, -34.8] p n/a P(B better) 0.94
MAPPO + flip vs Single-agent xN + flip	-45.2 pp [-48.3, -42.1] p n/a P(B better) 1.00	-46.0 pp [-49.3, -42.7] p n/a P(B better) 0.99	-46.1 pp [-49.5, -42.6] p n/a P(B better) 1.00	-42.0 pp [-46.9, -37.1] p n/a P(B better) 0.94
MAPPO (last) vs Single-agent xN + flip	-39.3 pp [-42.7, -35.8] p n/a P(B better) 0.98	-43.9 pp [-46.9, -40.9] p n/a P(B better) 0.98	-46.6 pp [-49.8, -43.4] p n/a P(B better) 0.98	-45.6 pp [-50.1, -41.2] p n/a P(B better) 0.98
Single-agent fine-tuned xN + flip vs Sin	-49.5 pp [-52.4, -46.6] p n/a P(B better) 1.00	-49.0 pp [-52.0, -45.9] p n/a P(B better) 1.00	-43.2 pp [-50.3, -36.1] p n/a P(B better) 0.99	-23.0 pp [-47.5, +1.5] p n/a P(B better) 0.90
IPPO vs Single-agent xN + flip	-54.8 pp [-58.0, -51.6] p n/a P(B better) 1.00	-59.7 pp [-62.7, -56.7] p n/a P(B better) 1.00	-58.5 pp [-61.8, -55.3] p n/a P(B better) 1.00	-60.2 pp [-64.1, -56.2] p n/a P(B better) 0.99
IPPO + flip vs Single-agent xN + flip	-54.8 pp [-58.0, -51.6] p n/a P(B better) 1.00	-59.7 pp [-62.7, -56.7] p n/a P(B better) 1.00	-58.5 pp [-61.7, -55.3] p n/a P(B better) 1.00	-60.1 pp [-64.1, -56.2] p n/a P(B better) 0.99
IPPO (last) vs Single-agent xN + flip	-54.8 pp [-58.0, -51.6] p n/a P(B better) 1.00	-59.7 pp [-62.7, -56.7] p n/a P(B better) 1.00	-58.5 pp [-61.8, -55.3] p n/a P(B better) 1.00	-60.2 pp [-64.1, -56.2] p n/a P(B better) 0.99

Success rate (99 %): percentage points, higher is better

B vs A	1 robots	2 robots	4 robots	8 robots
Single-agent xN + flip vs Single-agent x	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50	+89.7 pp [+85.8, +93.5] p n/a P(B better) 0.99	+98.3 pp [+96.9, +99.8] p n/a P(B better) 1.00	+100.0 pp [+100.0, +100.0] p n/a P(B better) 1.00
MAPPO vs Single-agent xN	+1.0 pp [-0.7, +2.7] p n/a P(B better) 0.52	+89.7 pp [+85.8, +93.5] p n/a P(B better) 0.99	+98.0 pp [+96.4, +99.6] p n/a P(B better) 1.00	+98.7 pp [+97.4, +100.0] p n/a P(B better) 1.00
MAPPO + flip vs Single-agent xN	+1.0 pp [-0.7, +2.7] p n/a P(B better) 0.52	+89.7 pp [+85.8, +93.5] p n/a P(B better) 0.99	+98.0 pp [+96.4, +99.6] p n/a P(B better) 1.00	+99.7 pp [+99.0, +100.3] p n/a P(B better) 1.00
MAPPO (last) vs Single-agent xN	+1.7 pp [+0.2, +3.1] p n/a P(B better) 0.53	+89.7 pp [+85.8, +93.5] p n/a P(B better) 0.99	+98.3 pp [+96.9, +99.8] p n/a P(B better) 1.00	+100.0 pp [+100.0, +100.0] p n/a P(B better) 1.00
Single-agent fine-tuned xN + flip vs Sin	+1.3 pp [+0.0, +2.6] p n/a P(B better) 0.52	+81.0 pp [+76.2, +85.8] p n/a P(B better) 0.99	+98.0 pp [+96.4, +99.6] p n/a P(B better) 1.00	+99.7 pp [+99.0, +100.3] p n/a P(B better) 1.00
IPPO vs Single-agent xN	+1.7 pp [+0.2, +3.1] p n/a P(B better) 0.53	+89.7 pp [+85.8, +93.5] p n/a P(B better) 0.99	+98.3 pp [+96.9, +99.8] p n/a P(B better) 1.00	+100.0 pp [+100.0, +100.0] p n/a P(B better) 1.00
IPPO + flip vs Single-agent xN	+1.7 pp [+0.2, +3.1] p n/a P(B better) 0.53	+89.7 pp [+85.8, +93.5] p n/a P(B better) 0.99	+98.3 pp [+96.9, +99.8] p n/a P(B better) 1.00	+100.0 pp [+100.0, +100.0] p n/a P(B better) 1.00
IPPO (last) vs Single-agent xN	+1.7 pp [+0.2, +3.1] p n/a P(B better) 0.53	+89.7 pp [+85.8, +93.5] p n/a P(B better) 0.99	+98.3 pp [+96.9, +99.8] p n/a P(B better) 1.00	+100.0 pp [+100.0, +100.0] p n/a P(B better) 1.00
MAPPO vs Single-agent xN + flip	+1.0 pp [-0.7, +2.7] p n/a P(B better) 0.52	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50	-0.3 pp [-1.0, +0.3] p n/a P(B better) 0.49	-1.3 pp [-2.6, -0.0] p n/a P(B better) 0.48
MAPPO + flip vs Single-agent xN + flip	+1.0 pp [-0.7, +2.7] p n/a P(B better) 0.52	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50	-0.3 pp [-1.0, +0.3] p n/a P(B better) 0.49	-0.3 pp [-1.0, +0.3] p n/a P(B better) 0.49
MAPPO (last) vs Single-agent xN + flip	+1.7 pp [+0.2, +3.1] p n/a P(B better) 0.53	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50
Single-agent fine-tuned xN + flip vs Sin	+1.3 pp [+0.0, +2.6] p n/a P(B better) 0.52	-8.7 pp [-11.6, -5.8] p n/a P(B better) 0.37	-0.3 pp [-1.0, +0.3] p n/a P(B better) 0.49	-0.3 pp [-1.0, +0.3] p n/a P(B better) 0.49
IPPO vs Single-agent xN + flip	+1.7 pp [+0.2, +3.1] p n/a P(B better) 0.53	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50
IPPO + flip vs Single-agent xN + flip	+1.7 pp [+0.2, +3.1] p n/a P(B better) 0.53	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50
IPPO (last) vs Single-agent xN + flip	+1.7 pp [+0.2, +3.1] p n/a P(B better) 0.53	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50	+0.0 pp [+0.0, +0.0] p n/a P(B better) 0.50

Collisions between robots per episode: difference, lower is better

B vs A	1 robots	2 robots	4 robots	8 robots
Single-agent xN + flip vs Single-agent x	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-13995.43 [-14575.56, -13415.30] p n/a P(B better	-30814.35 [-31260.08, -30368.62] p n/a P(B better	-63060.61 [-63166.25, -62954.97] p n/a P(B better) 1.00
MAPPO vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-14116.41 [-14702.72, -13530.10] p n/a P(B better	-31037.48 [-31484.71, -30590.26] p n/a P(B better	-63400.37 [-63540.22, -63260.51] p n/a P(B better) 1.00
MAPPO + flip vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-14116.41 [-14702.72, -13530.10] p n/a P(B better	-31037.98 [-31485.24, -30590.72] p n/a P(B better	-63497.05 [-63588.51, -63405.59] p n/a P(B better) 1.00
MAPPO (last) vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-14116.45 [-14702.78, -13530.12] p n/a P(B better	-31039.07 [-31486.43, -30591.72] p n/a P(B better	-63496.25 [-63589.32, -63403.18] p n/a P(B better) 1.00
Single-agent fine-tuned xN + flip vs Sin	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-13173.46 [-13790.98, -12555.94] p n/a P(B better	-30452.63 [-30925.46, -29979.80] p n/a P(B better	-62715.90 [-62955.04, -62476.77] p n/a P(B better) 1.00
IPPO vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-14116.50 [-14702.83, -13530.18] p n/a P(B better	-31039.36 [-31486.71, -30592.01] p n/a P(B better	-63504.43 [-63596.46, -63412.40] p n/a P(B better) 1.00
IPPO + flip vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-14116.50 [-14702.83, -13530.18] p n/a P(B better	-31039.36 [-31486.71, -30592.01] p n/a P(B better	-63504.43 [-63596.46, -63412.40] p n/a P(B better) 1.00
IPPO (last) vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-14116.50 [-14702.83, -13530.18] p n/a P(B better	-31039.36 [-31486.71, -30592.01] p n/a P(B better	-63504.43 [-63596.46, -63412.40] p n/a P(B better) 1.00
MAPPO vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-120.98 [-141.55, -100.41] p n/a P(B better) 1.00	-223.13 [-241.94, -204.32] p n/a P(B better) 0.99	-339.76 [-462.52, -216.99] p n/a P(B better) 0.97
MAPPO + flip vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-120.98 [-141.55, -100.41] p n/a P(B better) 1.00	-223.63 [-242.18, -205.07] p n/a P(B better) 1.00	-436.44 [-466.72, -406.16] p n/a P(B better) 1.00
MAPPO (last) vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-121.02 [-141.59, -100.45] p n/a P(B better) 1.00	-224.72 [-242.94, -206.50] p n/a P(B better) 1.00	-435.64 [-464.51, -406.77] p n/a P(B better) 1.00
Single-agent fine-tuned xN + flip vs Sin	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	+821.97 [+579.62, +1064.32] p n/a P(B better) 0.2	+361.72 [+168.56, +554.88] p n/a P(B better) 0.21	+344.71 [+122.25, +567.17] p n/a P(B better) 0.24
IPPO vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-121.07 [-141.64, -100.51] p n/a P(B better) 1.00	-225.01 [-243.22, -206.80] p n/a P(B better) 1.00	-443.82 [-472.60, -415.04] p n/a P(B better) 1.00
IPPO + flip vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-121.07 [-141.64, -100.51] p n/a P(B better) 1.00	-225.01 [-243.22, -206.80] p n/a P(B better) 1.00	-443.82 [-472.60, -415.04] p n/a P(B better) 1.00
IPPO (last) vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-121.07 [-141.64, -100.51] p n/a P(B better) 1.00	-225.01 [-243.22, -206.80] p n/a P(B better) 1.00	-443.82 [-472.60, -415.04] p n/a P(B better) 1.00

Workload imbalance (CV): difference, lower is better

B vs A	1 robots	2 robots	4 robots	8 robots
Single-agent xN + flip vs Single-agent x	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.01 [-0.02, +0.01] p n/a P(B better) 0.61	-0.48 [-0.53, -0.43] p n/a P(B better) 0.99	-0.50 [-0.55, -0.45] p n/a P(B better) 1.00
MAPPO vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.02 [-0.03, -0.01] p n/a P(B better) 0.67	-0.52 [-0.57, -0.47] p n/a P(B better) 0.99	-0.54 [-0.59, -0.49] p n/a P(B better) 1.00
MAPPO + flip vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.02 [-0.03, -0.01] p n/a P(B better) 0.67	-0.52 [-0.57, -0.47] p n/a P(B better) 0.99	-0.54 [-0.59, -0.49] p n/a P(B better) 1.00
MAPPO (last) vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.01 [-0.03, -0.00] p n/a P(B better) 0.58	-0.52 [-0.57, -0.48] p n/a P(B better) 0.99	-0.57 [-0.62, -0.52] p n/a P(B better) 1.00
Single-agent fine-tuned xN + flip vs Sin	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.00 [-0.02, +0.01] p n/a P(B better) 0.53	-0.41 [-0.46, -0.36] p n/a P(B better) 0.95	-0.43 [-0.48, -0.38] p n/a P(B better) 0.98
IPPO vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.02 [-0.03, -0.01] p n/a P(B better) 0.62	-0.53 [-0.58, -0.49] p n/a P(B better) 1.00	-0.56 [-0.61, -0.51] p n/a P(B better) 1.00
IPPO + flip vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.02 [-0.03, -0.01] p n/a P(B better) 0.62	-0.53 [-0.58, -0.49] p n/a P(B better) 1.00	-0.56 [-0.61, -0.51] p n/a P(B better) 1.00
IPPO (last) vs Single-agent xN	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.02 [-0.03, -0.01] p n/a P(B better) 0.62	-0.53 [-0.58, -0.49] p n/a P(B better) 1.00	-0.56 [-0.61, -0.51] p n/a P(B better) 1.00
MAPPO vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.01 [-0.02, +0.00] p n/a P(B better) 0.53	-0.04 [-0.05, -0.02] p n/a P(B better) 0.71	-0.04 [-0.05, -0.02] p n/a P(B better) 0.73
MAPPO + flip vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.01 [-0.02, +0.00] p n/a P(B better) 0.53	-0.04 [-0.05, -0.02] p n/a P(B better) 0.71	-0.04 [-0.05, -0.02] p n/a P(B better) 0.74
MAPPO (last) vs Single-agent xN + flip	+0.00 [+0.00, +0.00] p n/a P(B better) 0.50	-0.01 [-0.02, +0.01] p n/a P(B better) 0.47	-0.04 [-0.06, -0.03] p n/a P(B better) 0.72	-0.06 [-0.08, -0.05] p n/a P(B better) 0.83